# Can Satellites Replace the Field Plot? A Blind Benchmark of Satellite Forest Carbon Against Standard-Formula Ground Truth, and Conformal Crediting with Calibrated Over-Crediting Risk


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***Abstract*—Satellite platforms promise to replace field inventories in forest carbon crediting, but their estimates are rarely tested blind against the standard-formula answer, and credits are issued with deductions whose risk is unknown. We separate the question into two isolated grids. The right grid computes field ground truth to SNI 7724, IPCC, CDM, and Verra rules with a Monte Carlo uncertainty budget and is sealed. The left grid sees only plot coordinates and dates and estimates carbon from time-matched Sentinel-1/2, Landsat, PALSAR, and terrain data. A hash-chained ledger makes changes after the reveal detectable. On 476 stock plots, blind error was 29% on 1 ha plots with Sentinel, Landsat, and PALSAR, and 21% after adding a 1 m canopy height map and GEDI lidar. Project totals came within 7–9% outside Indonesia, while small Indonesian plots stayed near 100%. Counting trees on the 1 m canopy height map and applying the field formula found 3% of the stems and 25% of the carbon, and in 1,000 simulated projects the best map found a median 95% of the field carbon. Satellite embeddings and a regional bias correction raised the share of projects within $\pm 25\%$ of the field from 80% to 88.5%, and a region-level calibration held the credit's over-crediting rate at 9.7%; frozen and tested blind on 35 unseen Bornean plots, the method found 85% of the field carbon and its credit never exceeded it. Only 21.5% of Sentinel-2 acquisitions see an Indonesian plot clear, and a common scene-level filter discards 65% of those. We then propose conformal crediting: a project is credited a lower bound calibrated on other projects with field truth. Under cross-region transfer it met a 10% over-crediting target for each of five maps, within 2.9 points. It was invariant to map inflation, and credited 72–84% of the true stock without a field plot. The VM0047 deduction over-credited 23–29% of projects with field plots and sits below the integrity-yield frontier. For annual growth on 196 Bornean plots, no satellite estimate beat assuming no change. Satellites can map stock and support calibrated crediting; growth still needs remeasurement.**



***Index Terms*—forest carbon, MRV, SNI 7724, VM0047, remote sensing, cloud cover, Sentinel, ALOS PALSAR, machine learning, spatial cross-validation, conformal prediction, uncertainty, model-assisted estimation, commit-reveal ledger, distributed processing**


## I. Introduction

### A. Two Ways to Measure Forest Carbon

*Traditional MRV* (measurement, reporting, and verification) is field based. Crews lay out plots, measure each stem's diameter $D$ and height $H$, identify species to assign wood density $\rho$, and pass these inputs through a published allometric equation [1]. Carbon follows from a fixed carbon fraction and the $CO_2$ to C mass ratio. Methodologies such as Verra VM0047 prescribe this chain [2]; its output is deterministic and is the number on which credits are issued. Nobody can change the formula: it is the international standard. What is costly is the visit, especially in remote Indonesian peat swamps and hill forests.

*System-based MRV* replaces or supplements the visit with satellites and software. Optical sensors observe canopy greenness, radar observes canopy structure and moisture, and spaceborne lidar measures height [3]. Global biomass maps exist [4], [5], and commercial platforms, including our own NusaCarbon prototype, turn polygons into carbon numbers. The open question is how close such numbers are to what the field formula would have produced at the same place and time.

### B. Why the Comparison Is Hard to Do Honestly

Three traps inflate reported accuracy. First, *leakage*: if a model is trained or tuned on the plots it is scored on, or on plots next to them, the score measures memory rather than skill; spatially blocked validation of large-scale biomass

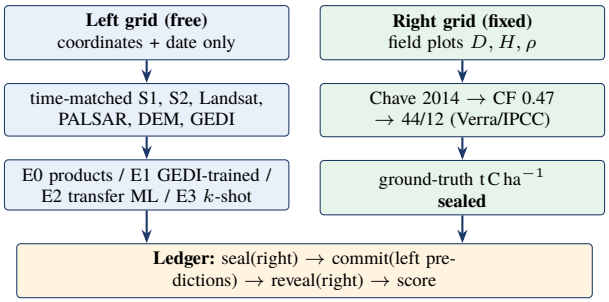



Fig. 1. Blind two-grid design. Only the public manifest (plot id, coordinates, date, radius) crosses from right to left.

maps reveals much lower skill than random validation [6], [7]. Second, *temporal mismatch*: a plot measured in 2009 and an image from 2023 describe different forests. Third, *scale*: a plot is a fraction of a hectare while satellite pixels range from 10 m to 4.6 km. A fair test must isolate the estimator from the answer, match time explicitly, and report how error changes with scale.

Our own platform illustrates the need. An audit of the NusaCarbon code base found that its forest and mangrove figures depended mostly on default constants (for example, a fixed sequestration rate and tree density bins from NDVI) rather than on satellite measurement, that clouds were screened only at scene level, that the reported confidence was a grade computed from image metadata, and that no estimate had been compared with field data. This paper builds the comparison that was missing, and tests the cloud handling and the confidence statement as carefully as the estimate itself.

### C. Two Isolated Grids

We separate the problem into two grids (Fig. 1). The **right grid** is fixed: published field plots, the standard formula, a carbon density in t C ha$^{-1}$. We never modify it. The **left grid** is free: any satellite data, any algorithm, any machine-learning method, subject to one rule: it never sees the right grid's outcomes for the plots it is scored on. A ledger enforces the rule and makes it auditable.

### D. Research Questions

**RQ1 (blind accuracy; research methodology).** How far does a blind, satellite-only estimate of aboveground carbon stock deviate from standard-formula field ground truth, and how does the deviation depend on the temporal gap between field and satellite observations and on the area over which it is aggregated?

**RQ2 (closing the gap; AI).** Which learning strategy brings the satellite estimate closest to the field answer: off-the-shelf global products, a model trained only on spaceborne lidar, a model transferred from other regions, or a model calibrated with $k$ local field plots? Are its prediction intervals honest? And can trees be counted from space, so that the field formula itself is applied to satellite-derived trees?

**RQ3 (clouds and confidence; AI and methodology).** How often do clouds hide an Indonesian plot from optical satellites, how much accuracy does that cost, and can the estimate and its prediction interval stay honest when they do? At the scale of a project, does the stated uncertainty protect against over-crediting, and how many field plots does a satellite save?

**RQ4 (sequestration).** Can the annual change in carbon stock, the quantity credits are issued on, be estimated blind from satellites, and how certain is the field answer itself?

**RQ5 (integrity; AI and methodology).** Can a credit be issued from a satellite estimate with a stated, verified risk of over-crediting, whatever map it comes from, and how many credits does that leave compared with the registry deduction and with field plots?

**RQ6 (distributed systems).** Can time-matched extraction for hundreds of plots and sensors be sharded, balanced, and made fault tolerant with bit-identical outputs, and can a hash-chained ledger make the order of prediction and reveal verifiable?

### E. Contributions

(i) A blind, time-matched evaluation protocol with a commit-reveal ledger, reusable for any MRV estimator. (ii) A right grid built step by step to SNI 7724, IPCC, CDM, Verra, and Gold Standard rules, with a Monte Carlo uncertainty budget, on 476 stock plots (148 Indonesian plots recomputed tree by tree) and 223 day-dated repeat-census plots in Borneo. (iii) An estimator ladder from zero field data to $k$ local plots, with spatially blocked validation. (iv) A plot-level Sentinel-2 clear-sky climatology for Indonesian forest plots and a measured error rate of the scene-level cloud filter. (v) A test of missing-data-augmented conformal intervals [8] under cloud-driven, non-random loss of the optical block with out-of-region calibration. (vi) An empirical coverage test of project-level intervals under spatial transfer, and a simulation of how often the VM0047 uncertainty deduction still over-credits. (vii) *Conformal crediting*: a crediting rule calibrated across projects against field truth. It meets its over-crediting target under spatial transfer and for biased maps, it is invariant to uniform map inflation, and it comes with an integrity-yield frontier that prices a field plot in credits. (viii) A tree-by-tree satellite estimator (E4) that counts trees on a 1 m canopy height map and applies the field formula, benchmarked against field stem lists and a wood-density classification of the field stems, and 1,000 project simulations reported as carbon amounts, with every plot row (Appendix A). (ix) Fixes for each limit found (satellite embeddings, a 2026 canopy height map, a realm bias correction, region-level credit calibration, an audit of a deployed height formula), frozen into one method and tested blind on 35 unseen plots (Section IV-G). (x) A sharded, idempotent extraction runner with measured scaling and fault tolerance.

**What is not new.** We propose no new estimator or interval method; the one methodological proposal, conformal crediting, reuses standard conformal machinery for a new decision. Conformalized quantile regression [9], its missing-data augmentation [8], modality dropout [10], model-assisted

estimation [11], [12], and the effects of plot size, saturation, allometric choice, and root-to-shoot choice [13]–[16] are established. Hash chains, sealed test sets, and commit-reveal are decades old [17]–[19]. The contribution is a blind, standards-anchored evaluation that tests these tools together on Indonesian and Bornean plots, under the failure modes an MRV platform meets: clouds, small plots, spatial transfer, and the deduction rules of the registries.

## II. Related Work

### A. Allometry and Standards

Pantropical allometry links $D$, $H$, and $\rho$ to aboveground biomass (AGB) [1]; the BIOMASS package propagates its uncertainty [20]. Verra VM0047 and the HCSA toolkit specify sampling, measurement thresholds, and plot designs [2], [21]. Investigations of REDD+ crediting show that unverifiable baselines produce over-crediting [22], which motivates auditable estimates.

### B. Remote Sensing of Biomass

Optical reflectance saturates over dense canopies. C-band radar mostly sees the upper canopy; L-band radar penetrates further and tracks woody biomass up to moderate densities [23], but its backscatter can *decrease* in very dense forest [24]. Spaceborne lidar from GEDI samples canopy height in 25 m footprints and is converted to biomass by footprint models calibrated on field data [3], [25]. Global AGB maps combine these signals: ESA CCI Biomass [4], the 2010 harmonized map of Spawn *et al.* [5], and the annual Chloris product [26]. Canopy height alone predicts plot carbon well where lidar is available [27].

### C. Machine Learning and Honest Validation

Gradient-boosted trees are strong tabular regressors [28]; models fitted on log biomass need a retransformation correction [29]. Spatial autocorrelation makes random cross-validation optimistic; blocked or leave-region-out designs are required [6], [7]. Conformal prediction yields distribution-free intervals with finite-sample coverage [30]. Agreement is assessed with RMSE, bias, and Lin's concordance [31].

### D. Clouds and Uncertainty

Southeast Asia is cloudy on at least 80% of days [32], and scene metadata suggest at most one cloud-free Sentinel-2 scene per year over much of tropical Indonesia [33]. The combined Sentinel-2 and Landsat revisit is 2.9 days [34]. Cloud masks such as Fmask and Cloud Score+ reach about 90% accuracy, with thin cloud and shadow the main failures [35]–[37]. Flores-Anderson *et al.* map the frequency of clear 30 m observations across the tropics [38]. Workarounds include best-available-pixel compositing [39], harmonized Landsat–Sentinel data [40], SAR–optical fusion [41], and deep SAR-to-optical cloud removal [42]. On the uncertainty side, split conformal prediction [43], conformalized quantile regression [9], normalized [44] and Mondrian [45] variants, and weighted or spatial conformal prediction under shift [46], [47] give distribution-free intervals; Zaffran *et al.* extend them to missing covariates by augmenting the calibration set with the test point's missingness pattern [8]. Conformal intervals have been applied to forest-attribute maps from Sentinel-2 [48] and to Earth-observation models in general [49]. For biomass maps, the CEOS protocol [50] and Duncanson *et al.* [51] recommend plots of at least 1 ha and explicit error propagation, and Araza *et al.* separate map error from plot error [14]. Model-assisted and model-based estimators that combine remote sensing with field samples are standard in forest inventory [11], [12], [52]; their efficiency grows with plot size [53], and model-based variance estimators under-cover when residual spatial correlation or model bias is ignored [54], [55].

### E. Integrity and Distributed Processing

Hash-chained time-stamping makes a document's existence at a time verifiable [17]; commit-reveal protocols bind a party to a value before another party reveals theirs [18]. Large geospatial workloads are processed as independent map tasks [56]; consistent hashing places keys on workers with minimal remapping [57], and Graham's longest-processing-time rule bounds makespan within $4/3$ of optimal [58]. Cloud catalogues such as the Microsoft Planetary Computer [59] and Google Earth Engine expose these archives through APIs.

## III. Methodology

### A. Right Grid: Field Ground Truth by the Standards

The right grid is the number a carbon standard would accept. We therefore build it the way the standards prescribe, from sampling design to deduction, and keep the deterministic point value separate from its uncertainty. Table I lists the plot sources and Table II the frameworks we follow.

**What a ground-truth plot is.** The coordinates of a plot only say where it is; the ground truth is the list of trees measured inside it. A field crew navigates to the coordinates, lays out the plot around the centre (for HCSA a 0.05 ha circle with a 0.01 ha inner circle for small stems), and tags every stem above the minimum diameter. For each stem it measures the diameter $D$ at 1.3 m with a diameter tape and the height $H$ with a clinometer or laser hypsometer, and it records the local name, which a botanist resolves to species, genus, and family. Wood density $\rho$, the oven-dry mass of wood per green volume, is a property of the species and is looked up in a wood density database by species, falling back to the genus or family mean. It matters because two stems of the same size can differ greatly in mass: in HCSA, *Campnosperma auriculata* has $\rho = 0.44$ and *Combretocarpus rotundatus* $0.70\,\mathrm{g\,cm^{-3}}$, so the second holds 1.6 times the biomass at the same $D$ and $H$. HCSA publishes the full list: 2,528 stem records of 370 species in 89 families, each with $D$, $H$, local and scientific name, family, and the $\rho$ assigned by the data provider from the ICRAF wood density database [60]. GEO-TREES and Labrière publish plot biomass that the networks computed from their own stem lists. None of these lists reaches the left grid, which receives only the manifest.

TABLE I
RIGHT-GRID FIELD DATASETS (STOCK)

| Source | Region | Plots | Blocks | Date basis |
|---|---|---|---|---|
| HCSA [60] | Indonesia (6 prov.) | 148 | 6 | release 2023 |
| GEO-TREES [61] | Pantropical | 142 | 15 | census year |
| Labrière [62] | Fr. Guiana, Gabon | 186 | 6 | campaign year |
| **Total** | | **476** | **27** | |

**Step 1, tree.** Aboveground biomass of a tree with diameter $D$ (cm), height $H$ (m), and wood density $\rho$ ($\mathrm{g\,cm^{-3}}$) follows Chave 2014 Eq. 4 [1],

$$\mathrm{AGB}_i = 0.0673\,(\rho_i D_i^2 H_i)^{0.976}\ \text{[kg]}, \qquad \sigma_{\ln} = 0.357, \quad (1)$$

where $\sigma_{\ln}$ is the residual standard error on the log scale. Where height is not measured, a local model $\ln H = a + b \ln D$ with Duan correction replaces it (SAFE: $a = 0.549$, $b = 0.656$, residual SD 0.348); Chave Eq. 7 with the environmental stress term $E$ is the fallback where no local heights exist. For HCSA, wood density is the per-stem value of the archive, and the level at which the Global Wood Density Database [71], [72] resolves each stem (species, genus, or family) sets its Monte Carlo error. For SAFE, wood density is the database mean at the finest level available, preferring South-East Asian records. As a check on model choice we also apply the East-Kalimantan equation of Basuki *et al.* [73], $\ln \mathrm{AGB} = -0.744 + 2.188 \ln D + 0.832 \ln \rho$ (correction factor 1.047). Palms are excluded: Chave's equations are for dicotyledonous trees, and palm allometry depends on stem height, not diameter [74], [75].

**Step 2, plot.** Tree biomass is expanded by the area in which a stem of its size is counted. For the HCSA nested design [21],

$$\mathrm{AGB}_j = \sum_{i \in j} \frac{\mathrm{AGB}_i}{1000\, a(D_i)}, \quad a(D) = \begin{cases} 0.01, & 5 \le D < 15 \\ 0.05, & D \ge 15 \end{cases} \quad (2)$$

in $\mathrm{Mg\,ha^{-1}}$; SNI 7724 uses the same form, $C_n = C_x/1000 \times 10{,}000/l_{\text{plot}}$, with its own subplot areas [63].

**Step 3, pools and carbon.** The plot carbon density is

$$\begin{aligned} C_j &= \mathrm{CF}\,(1+R)\,\mathrm{AGB}_j + C_{\mathrm{DW},j} + C_{\mathrm{L},j} + C_{\mathrm{SOC},j}, \\ \mathrm{CO_2e}_j &= \tfrac{44}{12}\, C_j, \end{aligned} \quad (3)$$

with $\mathrm{CF} = 0.47$ (IPCC 2006 Table 4.3, unchanged in 2019; SNI 7724; Verra) or 0.475 (Gold Standard). For the root-to-shoot ratio $R$ the frameworks disagree: SNI 7724 Lampiran A adopts the IPCC 2006 tropical-rainforest value $R = 0.37$, while the 2019 Refinement gives 0.207 (SD 0.072) below $125\,\mathrm{Mg\,ha^{-1}}$ and 0.212 (SD 0.077) above for natural Asian tropical rainforest [65], [66]. Dead wood, litter, and soil carbon are not in the public records; omitting a pool can only lower a stock claim, which is the conservative direction. All comparisons with the left grid use AGB, which removes the $R$ and CF choices from the benchmark; we report their effect separately.

**Step 4, stratum and project.** Following AR-TOOL14 Eqs. 14–17 [68], plots are grouped into strata $h$ (the HCS classes) with weights $w_h$, and the project mean and its relative uncertainty at 90% are

$$\bar{b} = \sum_h w_h \bar{b}_h, \qquad u = \frac{t_{0.95,\,n-M}\sqrt{\sum_h w_h^2 s_h^2 / n_h}}{\bar{b}}, \quad (4)$$

with $s_h^2$ the within-stratum variance, $n_h$ plots in stratum $h$, $n$ plots in total, and $M$ strata. The mapped stratum areas are not public, so the plot shares stand in for $w_h$. The number of plots needed for a margin $E$ (AR-TOOL03 Eq. 2, Neyman allocation, iterated on $t$ when $n < 30$) [69] is

$$n = \left(\frac{t_{0.95}}{E}\right)^2 \Big(\sum_h w_h s_h\Big)^2, \qquad n_h = n\,\frac{w_h s_h}{\sum_h w_h s_h}. \quad (5)$$

**Step 5, uncertainty.** AR-TOOL14 counts sampling error only. The IPCC asks for all sources, combined by Approach 1, $U = \sqrt{\sum_k U_k^2}$ for products and $U = \sqrt{\sum_k (U_k x_k)^2}/|\sum_k x_k|$ for sums, or by Monte Carlo (Approach 2) [67], which suits errors that are large, not normal, or combined nonlinearly. All three hold here: the allometric residual alone has $\sigma_{\ln} = 0.357$ per tree, wood density errors are skewed, and every error passes through the power law of Eq. 1, where summing variances is inexact. We use Approach 2 with 2,000 draws per tree and the error models of the BIOMASS package [20]: diameter error with SD $0.0062D + 0.0904$ cm for 95% of stems and 4.64 cm for a random 5% [76]; height error of 10% [76]; wood-density SD of 0.0708, 0.0941, and $0.1234\,\mathrm{g\,cm^{-3}}$ at species, genus, and family level; the allometric residual $\varepsilon_i \sim \mathcal{N}(0, \sigma_{\ln}^2)$; $\mathrm{CF} \sim \mathcal{N}(0.4713, 0.0206^2)$ [77]; and $R$ with its 2019 SD. The project error budget splits the variance of $\bar{b}$ into a sampling part (Eq. 4) and a measurement-plus-allometry part (the Monte Carlo variance of the same weighted mean).

**Step 6, change and deduction.** Sequestration uses the IPCC stock-difference method, $\Delta C = (C_{t_2} - C_{t_1})/(t_2 - t_1)$ (Vol. 4 Eq. 2.5), on the same plots at both dates [65]. Its uncertainty follows VM0047 v1.1 Eq. 28 [2],

$$\begin{aligned} U &= \frac{T\sqrt{SE_0^2 + SE_1^2 - 2\rho\, SE_0 SE_1}}{\Delta C}, \\ \mathrm{UNC} &= \min\big(1, \max(0,\ U - 0.10)\big), \end{aligned} \quad (6)$$

where $SE_0$ and $SE_1$ are the standard errors of the mean stock at the two dates, $\rho$ is their correlation on permanent plots, and $T$ is the two-tailed $t$ value at $\alpha = 0.1$. Credits are multiplied by $1 - \mathrm{UNC}$. For stocks, we report the deductions of VM0047 ($u - 10\%$), AR-TOOL14 (a stepped share of $u$: 0 up to 10%, then 25, 50, 75, 100% of $u$ at 15, 20, 30%), Gold Standard ($u - 20\%$), and VMD0017 (95% uncertainty minus 15%) [68], [70], [78]. In the Monte Carlo for change, a tree keeps the same allometric and height residual at both censuses, because the same stem is measured with the same equation; we also run the case in which they are redrawn.

**Data screening.** For HCSA, Step 1 is recomputed for every tree. Sixty-two records stated an area that contradicts Eq. 2 and were expanded by the protocol area. Whole plots were excluded, never imputed, if any record had a missing input, $H < 1.3$ m, conflicting coordinates, or $D > 200$ cm. The last

TABLE II
FIELD-INVENTORY RULES OF THE FRAMEWORKS APPLIED TO THE RIGHT GRID

| Framework | Plot design (minimum stem size) | Confidence | Precision | Uncertainty rule |
|---|---|---|---|---|
| SNI 7724:2011 [63] | nested 2×2 (seedlings, litter), 5×5 (saplings 2–10 cm), 10×10 (poles 10–20 cm), 20×20 m (trees ≥20 cm); five pools, Tier 3 | not stated | sampling error ≤20% | stratified systematic or random sampling |
| Indonesian NFI [64] | 20 km grid; 100×100 m permanent plot of 16 units of 25×25 m; subplot radii 1/2/5 m; trees ≥20 cm | – | – | national FRL, Manuri allometry, CF 0.47 |
| HCSA Toolkit v2.0 [21] | circular, $r = 5.64$ m for 5–14.9 cm, $r = 12.61$ m (0.05 ha) for ≥15 cm | 90% | ±10% | classes HDF, MDF, LDF, YRF, S, OL |
| IPCC 2006/2019 [65]–[67] | five pools; CF 0.47; root-to-shoot $R$ by zone | 95% | – | Approach 1 propagation or Approach 2 Monte Carlo |
| CDM AR-TOOL14/03 [68], [69] | stratified random plots | 90% | ±10% | stepped discount above 10%; sampling error only |
| Verra VM0047 v1.1 [2] | plot sampling, remeasured ≤5 yr | 90% | – | deduct $U - 10\%$; zero credit if $U > 100\%$ |
| Gold Standard A/R [70] | Winrock sourcebook plots; CF 0.475 | 90% | ±20% | deduct the excess over 20% |

rule caught plot RU09 (Riau), where a *Gluta renghas* was recorded at $D = 480$ cm, which matches a circumference typed into the diameter column ($480/\pi \approx 153$ cm); left in place it would have produced 4,426 Mg ha$^{-1}$. Of 185 plots, 148 were retained; the 15 records labelled as oil palm all fell in excluded plots. GEO-TREES publishes plot AGB from forest networks [61]; 154 plots with withheld AGB were dropped. Labrière *et al.* publish 1 ha units with AGB from Chave Eq. 4 and local height-diameter models, plus 95% credible intervals [62]; we use the 1 ha units only.

The outcome table is written to a sealed file. Only a manifest of plot id, region block, coordinates, date, date precision, and extraction radius is released to the left grid.

### B. Temporal Matching

Field inventories record a date or, more often, a year; no public inventory records the minute of measurement, and optical satellites revisit every 5–16 days with frequent cloud. We therefore match by window: for a plot measured at $t_j$, the left grid uses only clear observations in $[t_j - W,\ t_j + W]$, records the offset in days of the closest one, and takes the median of up to three clear scenes. The default is $W = 90$ days; RQ1 varies $W \in \{30, 90, 180, 365\}$ on the Indonesian plots. Each record also carries its timestamp precision (*day*, *year*, or *campaign year*) so that the analysis can separate sensor noise from date uncertainty.

### C. Left Grid: Satellite Features

For each plot the left grid reads, inside a square window around the plot centre: Sentinel-2 L2A reflectance in ten bands with scene-classification cloud masking and seven indices (NDVI, EVI, NDRE, NDMI, NBR, IRECI, SAVI); Landsat 5–9 Collection 2 surface reflectance with QA cloud masking; Sentinel-1 RTC backscatter (VV, VH, ratio); the ALOS PALSAR annual L-band mosaic (HH, HV); and Copernicus 30 m elevation and slope. Two global products are read as zero-shot baselines: Chloris annual AGB (2003–2019) [26] and the 2010 harmonized map [5]. Through Google Earth Engine the left grid also reads the Meta 1 m [79] and ETH 10 m [80] canopy height maps, in the plot and in a 50 m buffer. It also reads GEDI L4A footprint biomass and L2A relative height within 500 m and 1 km (quality-filtered, April 2019 to March 2023), the GEDI L4B 1 km grid [25], [81], ESA CCI Biomass for the nearest year [4], the ALOS/ALOS-2 PALSAR mosaic for the nearest year (2007–2024), and Hansen forest loss up to the field year [82]. The canopy height maps describe 2018–2020 imagery and GEDI 2019–2023, so for plots measured earlier they are not time-matched.

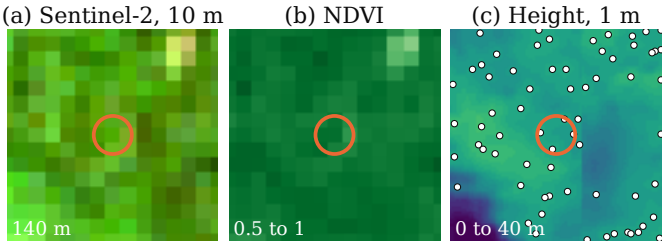



Fig. 2. What the left grid sees around HCSA plot KT30, Central Kalimantan (field: 342 Mg ha$^{-1}$ from 23 stems; circle: the 0.05 ha plot), on one 140 m window: (a) Sentinel-2 true colour at 10 m (2 September 2023, every pixel clear); (b) its NDVI, 0.88–0.95 (5th–95th percentile) across the window; (c) the 1 m canopy height (0–40 m) with the 67 tree tops E4 finds, four inside the plot.

**From pixels to carbon.** A satellite does not see trees (Fig. 2). It records, per pixel, the light reflected in each band and the radar echo; GEDI records the height profile of 25 m footprints. Optical bands are combined into vegetation indices, of which the best known is the normalized difference vegetation index [83],

$$\mathrm{NDVI} = \frac{R_{\mathrm{NIR}} - R_{\mathrm{red}}}{R_{\mathrm{NIR}} + R_{\mathrm{red}}}, \tag{7}$$

with $R$ the surface reflectance in Sentinel-2 band B8 (842 nm) and B4 (665 nm), or Landsat bands 4 and 3 (TM) and 5 and 4 (OLI). Chlorophyll absorbs red light for photosynthesis and the

**Algorithm 1** Satellite to Carbon for One Plot (E2)

**Require:** plot coordinates, field date $t$, radius $r$ (the manifest)
**for** each sensor (Sentinel-2, Landsat, Sentinel-1, PALSAR) **do**
  find scenes within $t \pm 90$ days
  keep a scene only if every pixel of the plot window is clear
  window means of bands; indices such as NDVI (Eq. 7)
  median over up to three clear scenes
**end for**
add terrain, canopy height percentiles, GEDI biomass and height
$\hat{y} \leftarrow f_{-R}(x)$, LightGBM on ln AGB trained on the plots of every other region
$\widehat{\mathrm{AGB}} \leftarrow e^{\hat{y}} \cdot \overline{e^{\varepsilon}}$ (Duan smearing)
**return** $C = 0.47\,\widehat{\mathrm{AGB}}$, $\mathrm{CO_2e} = \frac{44}{12} C$ per hectare

cell structure of leaves reflects near-infrared, so a green canopy reads about 0.8, bare soil 0.1–0.2, and water below zero. NDVI measures how green a pixel is, not how much wood stands under the leaves; Section IV-E shows that it saturates in closed forest. Algorithm 1 gives the whole chain from pixels to carbon for one plot.

### D. Clouds

Southeast Asia is cloudy on at least 80% of days, with no reliable clear season [32], and Sentinel-2 metadata for 2017 showed at most one cloud-free scene over much of tropical Indonesia [33]. Tropical maps of clear-observation frequency exist at 30 m [38], but we found no Sentinel-2 statistic at Indonesian plot scale and no measurement of how often scene-level filters misjudge a plot, so we measured both, then designed the estimator so that clouds cost as little as possible.

**C1, plot-level climatology.** For 97 locations (one per 0.1° cell holding a plot), we read the Sentinel-2 scene classification (SCL) of every acquisition from 2021 to 2023 in a 5×5 window of 20 m pixels around the plot: 22,697 acquisitions after duplicates of the same date were merged. An acquisition is *clear* if every pixel is vegetation, bare soil, water, or unclassified. From this we estimate the per-acquisition clear probability $p$, its monthly cycle, the gaps between clear looks, and the probability of at least one clear look within $\pm W$ days of a random date, which we compare with the independence model $1-(1-p)^m$ for $m$ acquisitions in the window; the difference measures how much clouds cluster in time. We also test the scene-level filter used by many Earth Engine pipelines, including our own platform (`CLOUDY_PIXEL_PERCENTAGE` $<30$ or $<40$), against the plot condition.

**C2, missing optical data.** When clouds hide a plot, the optical block (Sentinel-2 and Landsat features) is missing. For a fraction $q \in \{0, 0.25, 0.5, 0.75, 1\}$ of each target region's plots we remove it and compare four strategies, all leave-one-region-out: *naive* (the full model, missing values routed down LightGBM's default branches); *dropout*, one model trained on the real rows plus copies with the optical block removed, the modality-dropout idea of Neverova *et al.* [10]; *switch*, a cloud-aware ensemble that uses a radar-and-terrain model when optical data are missing; and *impute*, optical features regressed from radar, a simple form of SAR-to-optical gap filling [42].

**Cloud-aware conformal intervals.** A prediction interval is only honest if it is calibrated on cases like the one being predicted. When clouds are rare in the calibration data, a group-conditional (Mondrian) calibration [45] of the cloud-hit group has too few scores. We therefore use the missing-data augmentation of Zaffran *et al.* [8]: every inner out-of-fold calibration plot is scored twice, as observed and with its optical block removed, with quantile models trained under modality dropout [10], and each test plot takes the threshold of its own missingness pattern. With two patterns this is a special case of their CP-MDA. Their guarantee assumes exchangeable data and missingness independent of the outcome; here clouds follow climate and region, and calibration is out-of-region, so neither holds and coverage has to be measured. We test it on every target plot, both as observed and fully clouded.

### E. Estimator Ladder

**E0 (off the shelf).** Global products read at the plot, no fitting. **E1 (no field data).** The published GEDI L4A and L4B biomass products, read as they are, serve as the estimator that uses no field plot of ours. A model trained on GEDI footprint labels is implemented but not evaluated. **E2 (transfer).** LightGBM on ln AGB, trained on all region blocks except the target block and predicting it blind (leave-one-region-out), with Duan smearing [29]. **E3 ($k$-shot).** E2 followed by a ratio calibration on $k$ randomly drawn plots of the target block, scored on its remaining plots, repeated 200 times per $k$. Split-conformal intervals for E2 are calibrated on inner leave-one-region-out residuals of the training blocks only [30].

**E4 (count the trees, then the field formula).** E2 learns a direct map from pixels to biomass. E4 instead rebuilds the right grid's own tree list from the 1 m canopy height map and applies the right grid's formula to it, so that the two grids differ only in where each tree's numbers come from (Algorithm 2). A 140×140 m patch of the Meta map is read on the local UTM grid at 1 m around each plot. The map stores whole metres, so it is smoothed by a Gaussian of 1 m. Tree tops are local maxima taller than 5 m in a window whose radius grows with height, $r = 0.06H + 0.5$ m, the variable window filter of Popescu and Wynne [84]. Each crown is flooded from its top, highest pixels first, while the canopy stays at least 2 m tall, at least half the top's height, and within 15 m. The stem diameter follows from height and crown diameter by Jucker *et al.* [85],

$$D = a\,(H \cdot CD)^b \text{ [cm]}, \tag{8}$$

with the tropical-angiosperm coefficients of the plot's realm: $(a, b) = (0.486, 0.802)$ for Indo-Malaya, $(0.428, 0.821)$ for the Afrotropic, and $(0.591, 0.781)$ for the Neotropic. A satellite cannot identify species, so every tree takes the mean wood density of the tropical species of its region in the Global Wood Density Database (each species counted once): 0.574 for South-East Asia, 0.615 for Africa, and 0.630 for South America. Eq. 1 then gives each tree's biomass, the trees whose top falls inside the plot footprint (0.05 ha for HCSA, 1 ha otherwise) are summed, and the sum is divided by the footprint area. E4 uses no field data at all. *E4 scaled* multiplies it by

**Algorithm 2** E4: Count the Trees, Then the Field Formula

**Require:** 1 m canopy height patch $h$, plot footprint, realm, region
1: $h \leftarrow$ Gaussian smoothing of $h$, $\sigma = 1$ m
2: tops $\leftarrow \{p : h(p) \geq 5,\ h(p) = \max_{|q-p| \leq 0.06h(p)+0.5} h(q)\}$
3: grow crowns from the tops, highest pixel first, while $h \geq \max(2,\ h_{\text{top}}/2)$ and within 15 m
4: **for** each top $i$ inside the footprint **do**
5: $H_i \leftarrow h(\text{top}_i)$, $CD_i \leftarrow 2\sqrt{A_i/\pi}$ ($A_i$ crown area)
6: $D_i \leftarrow a(H_i CD_i)^b$ (Eq. 8); drop the tree if $D_i < 5$ cm
7: $\text{AGB}_i \leftarrow 0.0673(\bar{\rho}\, D_i^2 H_i)^{0.976}$, $\bar{\rho}$ the regional default
8: **end for**
9: **return** $\sum_i \text{AGB}_i/(1000\, A_{\text{plot}})$ in Mg ha$^{-1}$, then $\times 0.47 \times \frac{44}{12}$

the ratio of field to E4 totals on all other regions, which is the least field data a correction can use. E4 was designed after the stock reveal and is exploratory; as a sensitivity we also run it without smoothing.

### *F. Confidence: Plot and Project Intervals*

**Plot intervals.** All intervals target 90% and are calibrated on inner leave-one-region-out residuals of the training regions only. With calibration scores $s_1, \ldots, s_n$, the conformal threshold is the $\lceil (n+1)(1-\alpha) \rceil$-th smallest score, which guarantees marginal coverage between $1-\alpha$ and $1-\alpha+1/(n+1)$ under exchangeability [43]. We compare four scores on $y = \ln \text{AGB}$:

$$\text{split:} \quad s = |y - \hat{y}|, \qquad \hat{C} = \hat{y} \pm q, \tag{9}$$

$$\text{Mondrian:} \quad \text{split, one } q_g \text{ per availability group } g, \tag{10}$$

$$\text{normalized:} \quad s = |y - \hat{y}|/\hat{\sigma}(x), \qquad \hat{C} = \hat{y} \pm q\,\hat{\sigma}(x), \tag{11}$$

$$\text{CQR:} \quad s = \max\{\hat{q}_{\text{lo}}(x) - y,\ y - \hat{q}_{\text{hi}}(x)\}, \tag{12}$$

$$\hat{C} = [\hat{q}_{\text{lo}}(x) - q,\ \hat{q}_{\text{hi}}(x) + q], \tag{13}$$

following Papadopoulos *et al.* [44] for the normalized score, with $\hat{\sigma}(x)$ a second LightGBM fitted to the inner residuals, and Romano *et al.* [9] for conformalized quantile regression (CQR), with LightGBM quantile models at 5% and 95%. We report coverage, the median ratio of the upper to the lower bound, and the Winkler interval score on the log scale, $(\hat{u} - \hat{l}) + \frac{2}{\alpha}\big[(\hat{l} - y)_+ + (y - \hat{u})_+\big]$. As Araza *et al.* note, an interval calibrated on plots covers the plot value, which includes field error, not the true stock [14].

**Is a metadata "confidence score" a confidence?** Monitoring platforms often grade their output with a score built from cloud cover, data completeness, and temporal offset. Ours did: 40 points for cloud cover, plus weights for completeness and stability. We rebuild such a score from the same inputs (40 points for clear optical scenes, 30 for the share of sensor groups present, 30 for temporal closeness) and test whether it ranks plots by their actual error, using Spearman's $\rho$ against $|\ln y - \ln \hat{y}|$, next to the learned $\hat{\sigma}(x)$, both raw and as a partial rank correlation that removes predicted AGB and the dataset or region, which could otherwise confound it.

**Project intervals.** Every region block with at least eight plots plays a project, whose true stock is the mean field AGB of its plots. For satellite-only estimates of the mean, a naive interval treats plot errors as independent, $SE = s_w/\sqrt{n}$. Map errors, however, are spatially correlated over 50–100 km [14], so a project shares one regional bias. The two-component interval adds the between-region variance of the region-mean log residual,

$$SE^2 = \frac{s_w^2}{n} + s_b^2, \tag{14}$$

with $s_w$ and $s_b$ estimated from the other regions only. When $k$ field plots are available, the satellite becomes auxiliary information for a model-assisted estimator [11], [52]. The ratio estimator is

$$\hat{\bar{Y}}_R = \frac{\sum_{i \in s} y_i}{\sum_{i \in s} \hat{y}_i}\, \bar{\hat{y}}, \qquad \widehat{V} = \Big(1 - \frac{k}{n}\Big)\frac{1}{k}\, s_e^2, \quad e_i = y_i - \hat{R}\hat{y}_i, \tag{15}$$

where $\bar{\hat{y}}$ is the satellite mean over the whole project. The regression (GREG) estimator is $\bar{y}_s + \hat{\beta}(\bar{\hat{y}} - \bar{\hat{y}}_s)$. Both are compared with the field-only mean of the same $k$ plots over 500 random draws. The relative efficiency $\text{MSE}_{\text{field}}/\text{MSE}_{\text{estimator}}$ says how many field plots one satellite-assisted plot is worth. We use the empirical MSE over draws, because the GREG variance formula underestimates at small $k$. A conservative estimate $\hat{Y}(1 - \text{UNC})$ with the VM0047 rule exceeds the truth whenever the project is over-credited; we report how often.

### *G. Conformal Crediting*

A registry does not issue an interval; it issues a number. The number should be a lower bound that exceeds the true stock in at most a stated share $\delta$ of projects. This is the conservativeness principle of Grassi *et al.* [86], which Neeff writes as $P\{\hat{Y}(1-d) \leq Y\} \geq 1 - \delta$ and solves parametrically with $d = t\, SE/\hat{Y}$ [87]. A parametric bound is only as good as its standard error. For a satellite map no design-based standard error exists, and the model-based one rests on an error model we have shown to be fragile. We therefore calibrate the bound on other projects whose field truth is known, as Singh *et al.* suggested in passing for registry intervals [49].

**Rule.** Let project $j$ have an estimate $\hat{Y}_j$ of its mean stock, true mean $Y_j$, and a relative standard error $r_j$ ($r_j = 1$ when none exists). Its score is $z_j = \ln(\hat{Y}_j/Y_j)/r_j$. For a new project, with scores $z_1, \ldots, z_m$ from $m$ calibration projects, the credited stock is

$$C = \hat{Y} \exp(-q\, r), \qquad q = z_{(\lceil (m+1)(1-\delta) \rceil)}, \tag{16}$$

the $\lceil (m+1)(1-\delta) \rceil$-th smallest score, or $C = 0$ if that rank exceeds $m$.

**Guarantee.** If the new project and the calibration projects are exchangeable, then $P(C > Y) \leq \delta$, and it is at least $\delta - 1/(m+1)$ when scores have no ties. This is split conformal prediction [43] with projects as the units. Plots inside a project may be correlated in any way, because each project contributes one score [88]. The bound needs $m \geq 1/\delta - 1$ calibration projects: 9 for $\delta = 0.1$ and 19 for $\delta = 0.05$.

**Invariance to map inflation.** If every satellite estimate is multiplied by $c > 0$, every plain score ($r = 1$) shifts by $\ln c$, so $q$ shifts by $\ln c$, and $C = c\hat{Y} \exp(-q - \ln c) = \hat{Y} \exp(-q)$ is unchanged. A developer who inflates a map uniformly gains no credit. A parametric bound $\hat{Y} \exp(-z_{1-\delta} SE)$ with $SE$ from residual spread grows by the full factor $c$.

**Variants and baselines.** The *studentized* variant uses the design-based relative standard error of the project estimate as $r_j$. With $k$ field plots this is the ratio estimator of Eq. 15, so the variant recalibrates the registry variance formula across projects. Breidenbach *et al.* found that such formulas undercover [54]. The *Mondrian* variant calibrates within the plot-size class, small (radius $\leq$15 m) or large, which is a public field of the manifest. Baselines are no deduction; the VM0047 deduction $\hat{Y}(1-\max(0, U-0.1))$; the design lower bound $\hat{Y}(1 - t_{1-\delta,k-1}r)$ of Neeff [87]; and the parametric satellite-only bounds with independent or two-component errors (Eq. 14).

**Integrity-constrained yield.** We score a rule by its realised over-crediting rate and by its *yield*, the mean of $C_j/Y_j$: the share of the true stock that is credited. Comparing designs by yield at a fixed realised risk gives the value of a field plot in credits.

**Experiment.** Projects are region blocks with at least 8 plots. Regions of 16 or more plots are split into spatial sub-projects of about 10 plots by $k$-means on coordinates, which gives 31 projects in 19 regions. A project is calibrated only on projects from other regions, so the guarantee is tested under transfer. With $k$ field plots, each project is drawn 200 times. Each draw is calibrated on one random draw per calibration project (the subsampling scheme for two-layer data [88]). Every run comes after the stock reveal, so the results are exploratory.

### H. Scoring

We report RMSE, relative RMSE, bias, $R^2$, and Lin's concordance [31] per dataset and overall, interval coverage for E2, error by field-AGB range, error versus window $W$, and the error of the mean of $n$ plots as $n$ grows, which approximates estimating a larger area.

### I. Sequestration

Credits are issued on change, not stock. The only public tropical dataset we found that gives repeated per-stem measurements with *day-precision* dates is the SAFE Project tree census in Sabah, Malaysian Borneo [89], [90]: 25 m$\times$25 m plots across old-growth, logged, riparian, and oil-palm land, censused from 2010 to 2020, with every stem's measurement date recorded. It is also the closest field site to Kalimantan in our benchmark.

The right grid applies the same chain to every census: Chave Eq. 4 on stems with $D \geq 10$ cm; height from a local $\ln H = a + b \ln D$ model fitted to the heights SAFE measured, with Duan correction; wood density from the Global Wood Density Database [71], [72] by species, then genus, then family, then plot mean, preferring South-East Asian records. For each plot with censuses at least three years apart, sequestration is

$$\Delta\mathrm{AGB}_j = \frac{\mathrm{AGB}_j(t_1) - \mathrm{AGB}_j(t_0)}{t_1 - t_0} \quad [\mathrm{Mg\,ha^{-1}\,yr^{-1}}], \quad (17)$$

where $t_0$ and $t_1$ are the median stem dates of the first and last census. This gives 223 plots in 18 blocks, with $t_0$ between 2010 and 2015 and $t_1$ between 2016 and 2020. The manifest released to the left grid holds only plot id, block, coordinates, and the two dates.

The left grid estimates $\Delta$AGB three ways: *no change* ($\Delta = 0$, the baseline any estimator must beat); *S0*, the difference of the Chloris annual maps between the census years; and *S2*, the difference of two E2 stock predictions at $t_0$ and $t_1$, with E2 trained on the 476 stock plots only. Sentinel-2 did not exist at most $t_0$ dates, so S2 uses Landsat, Sentinel-1, PALSAR, and terrain. No SAFE label is released to any estimator; the same E2 model also predicts SAFE stock at $t_0$, which is an out-of-region test on Borneo.

### J. Blindness Ledger

Every protocol step is an entry in an append-only log whose entries carry the SHA-256 of the previous entry [17]. An evaluation epoch is: `seal` (digest of the sealed outcomes and of the public manifest), `commit` (digest of each estimator's prediction file), `release` (for E2, the digest and size of each fold's training labels, with a flag stating whether any target-block label was included), and `reveal` (accepted only if the sealed file still matches its digest). Commits after a reveal are refused. Editing any entry breaks every later hash, so a third party can verify the order of events from the log alone [18].

### K. Distributed Extraction

One task is one (plot, date, window, sensor) request. Its result is stored under the SHA-256 of the canonical task, which makes re-execution idempotent and turns the cache into a checkpoint: a restarted run skips every finished task. Writes are atomic (write then rename). Transient errors are retried with capped exponential backoff and jitter. Tasks are grouped into spatial tiles so a worker reuses scenes for neighbouring plots, but field plots cluster (one site holds 89 plots), so a tile holding more than the per-worker mean is split into sub-buckets by plot id, keeping all sensors of a plot together. Buckets are then placed either by consistent hashing [57] or by the longest-processing-time rule [58]. On the 2,052 tasks of this study with 24 workers, the worst-worker load factor $\lambda = \max_w L_w/\bar{L}$ was 6.75 for tile-only placement, 2.18 for split tiles with consistent hashing, and 1.40 for split tiles with LPT.

## IV. Results

### A. How Certain Is the Right Grid?

Before scoring the satellites we score the field answer.

**Plot level.** On the 148 HCSA plots, the Monte Carlo 95% interval of plot AGB has a median half-width, half its length divided by the plot value, of 31% (IQR 25–45%): the field answer itself is known only to about $\pm$31%. Of the stems, 45% have a species-level wood density, 42% genus-level, 7% family-level, and 5% none. The Monte Carlo median is 5% above the deterministic value, because the allometric error is log-normal; the deterministic value is the one the standards use.

**Equation choice.** Consistent with earlier comparisons of generic and regional equations [15], the regional equation

TABLE III
RIGHT-GRID UNCERTAINTY PER PROVINCE (HCSA, AGB)

| Province | $n$ | $\bar{b}$ | $u_{90}$ (%) Sampl. | $u_{90}$ (%) Total | Meas. share | VM0047 deduct. | Plots for ±10% |
|---|---|---|---|---|---|---|---|
| C. Kalimantan | 33 | 250 | 13.8 | 14.0 | 10% | 3.8% | 36 |
| Jambi | 20 | 171 | 33.9 | 34.5 | 16% | 23.9% | 71 |
| Riau | 19 | 126 | 12.7 | 19.8 | 64% | 2.7% | 19 |
| S. Sumatra | 22 | 80 | 9.2 | 12.3 | 51% | 0.0% | 14 |
| W. Kalimantan | 36 | 165 | 14.4 | 15.1 | 15% | 4.4% | 23 |
| W. Papua | 18 | 116 | 8.8 | 12.4 | 57% | 0.0% | 8 |

$\bar{b}$: stratified mean AGB (Mg ha$^{-1}$). Meas. share: measurement and allometry share of the variance of $\bar{b}$.

TABLE IV
BLIND STOCK ACCURACY (MG AGB HA$^{-1}$; COMMITTED BEFORE REVEAL)

| Subset | Estimator | RMSE | rRMSE | Bias | $R^2$ | CCC |
|---|---|---|---|---|---|---|
| All (476) | E0 Chloris | 128.1 | 49.3% | −4.6% | 0.29 | 0.43 |
| | E0 HGB 2010 | 149.4 | 57.5% | −8.4% | 0.03 | 0.27 |
| | E2 LORO | 126.1 | 48.5% | −9.5% | 0.31 | 0.59 |
| GEO-T. (142) | E0 Chloris | 102.8 | 37.5% | −10.3% | 0.37 | 0.56 |
| | E2 LORO | 114.9 | 41.9% | 0.0% | 0.21 | 0.55 |
| HCSA (148) | E0 Chloris | 172.4 | 107.1% | +19.0% | −0.12 | 0.02 |
| | E2 LORO | 165.6 | 102.9% | −29.1% | −0.03 | 0.22 |
| Labr. (186) | E0 Chloris | 101.2 | 31.0% | −10.3% | 0.18 | 0.37 |
| | E2 LORO | 93.7 | 28.6% | −7.9% | 0.30 | 0.55 |

of Basuki *et al.* gives almost the same total over all plots (ratio 1.01). By region, however, the stock moves from −40% (Central Kalimantan) to +121% (South Sumatra), because it ignores height where HCSA measured short trees.

**Pool choice.** The root-to-shoot ratio of SNI 7724 (IPCC 2006) gives 13% more total carbon than the 2019 Refinement. This is close to the 12% shift in global vegetation carbon that Mokany *et al.* found from revising root-to-shoot ratios [16]. Two standards, both valid, differ by more than the precision target of most methodologies.

**Project level.** Table III applies AR-TOOL14 to each province. The sampling uncertainty at 90% is 9–34%, and five of six provinces meet SNI 7724's 20% limit. But measurement and allometric error, which AR-TOOL14 does not count, carry 10–64% of the variance of the project mean. Their share is largest where the plots vary least between each other (Riau, 64%). The deductions differ by standard for the same data (Jambi: 24% under VM0047, 34% under AR-TOOL14, 14% under Gold Standard). Reaching ±10% at 90% would take 8 to 71 plots per province.

**Change.** Growth is far harder than stock. On SAFE, the Monte Carlo SD of a plot's ΔAGB is 2.0 Mg ha$^{-1}$ yr$^{-1}$ when a tree keeps its allometric residual across censuses, against 4.3 when it is redrawn. The signal itself is 1.9. Remeasuring the same stems with the same equation cancels half the error. At block level, Eq. 6 gives 90% uncertainties of 101–901% when the correlation of the two stock means is ignored ($\rho = 0$), which is a 100% deduction for 13 of 15 blocks and over 90% for the other two. With the measured $\rho$ (0.45–0.99), the uncertainty falls to 28–469%, and seven blocks keep some credit, the best at an 18% deduction. With 9–16 plots of 0.06 ha per block, even the field standard cannot certify growth of 2 Mg ha$^{-1}$ yr$^{-1}$, unless it uses the correlation that permanent plots provide.

### B. Blind Accuracy of Carbon Stock (RQ1)

Table IV gives the blind scores. All three estimators were committed to the ledger before the reveal. Over all 476 plots the best blind estimate, E2, misses the field value by a relative RMSE of 48.5% and explains 31% of its variance; the Chloris product, which was never fitted to any of our plots, is almost as close (49.3%). The 2010 harmonized map is worse (57.5%, $R^2 = 0.03$), partly because it describes 2010 while most plots were measured years later. Because carbon is a fixed multiple of AGB on both grids, every relative figure applies unchanged to t C ha$^{-1}$ and t CO$_2$e ha$^{-1}$.

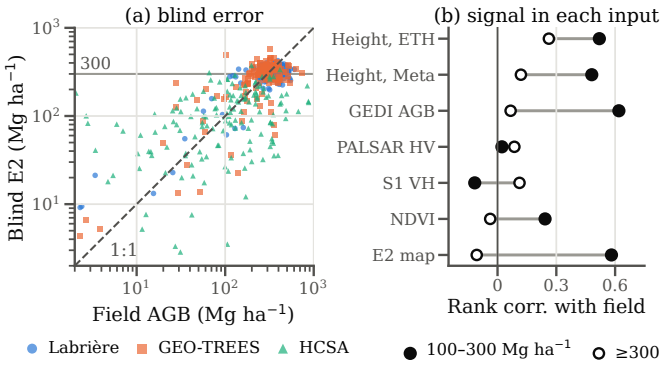



Fig. 3. Blind error and its source. (a) Blind E2 prediction against the field value for 475 plots (log axes): above about 300 Mg ha$^{-1}$ predictions flatten, and small HCSA plots scatter widely. (b) Spearman correlation of each input, and of E2, with field AGB in two bands of equal relative width (×3): 100–300 Mg ha$^{-1}$ (181 plots) and 300–876 (205). NDVI and Sentinel-1 have fewer plots (clear looks only).

The dataset matters more than the estimator. On the 1 ha Labrière units E2 is within 28.6% (CCC 0.55); on GEO-TREES within 41.9%; on the Indonesian HCSA plots every estimator misses by more than 100% and none beats predicting the mean ($R^2 < 0$).

### C. Where the Error Comes From

**Saturation.** Fig. 3a shows the pattern. Binned by field AGB, both E0 and E2 regress toward ∼300 Mg ha$^{-1}$. For plots above 400 Mg ha$^{-1}$ (mean 480) the E2 mean prediction is 293 and the Chloris mean is 295, a median under-estimate of about one third. Below 50 Mg ha$^{-1}$ (mean 21) E2 predicts 60 and Chloris 178. Between 100 and 300 Mg ha$^{-1}$ the median error of E2 is only +5 to +12%. The satellite signal therefore carries useful information in the middle of the range and little at either end, which is the behaviour reported for optical and L-band signals [23], [24], [91]. The cause is in the inputs, not the model (Fig. 3b). From field means of 262 to 480 Mg ha$^{-1}$ the median Meta canopy height rose only from 19.6 to 21.7 m, ETH from 31.8 to 37.1 m, GEDI biomass from 199 to 207 Mg ha$^{-1}$,

PALSAR HV by 0.2 dB, and NDVI not at all (0.83 to 0.81). Between 100 and 300 Mg ha$^{-1}$ canopy height and GEDI still rank plots (Spearman 0.48–0.62); above 300 none does (at most 0.26, ETH), and radar is flat already from 100. A model cannot separate plots its inputs do not, so it predicts the average of what it saw. Numbers: `input_saturation.csv` and `input_saturation_bins.csv`.

**Scale.** The HCSA plots are nested 0.01/0.05 ha plots with a median of 15 stems. The largest stem alone holds a median 24% of a plot's AGB (IQR 16–41%), and one stem of 1 m diameter adds about 110 Mg ha$^{-1}$ to a 0.05 ha plot (the largest single stem, 498), so one tree more or less moves a plot by tens to hundreds of Mg ha$^{-1}$, as found for small plots [13], [76]. Their field values span 2 to 876 Mg ha$^{-1}$ within one province. A 30 m pixel window cannot resolve that, and the 4.6 km Chloris pixel averages open land and forest together, which explains its large over-estimate on HCSA open-land and scrub plots. The HCSA land-cover labels would separate these plots well (within-class standard deviations of 10–40 Mg ha$^{-1}$ for all classes except high-density forest), but they are released with the field records, so they belong to the right grid and are never shown to the left grid. A satellite classifier that reproduced the HCS stratification is a promising left-grid estimator for future work.

**Field truth is not the bottleneck.** Labrière publish a 95% credible interval for every unit; the implied standard error of the field value is 5.0% (RMS), against a 28.6% satellite error on the same units. The right grid is far more precise than the left grid, so the gap is a satellite gap, not a ground-truth gap.

**Time.** Re-extracting the Indonesian plots' optical and C-band features with $W = 30$, 90, 180, and 365 days changed their blind error by less than three points (rRMSE 102.5, 104.8, 102.7, 102.0%), while the share of plots with at least one clear Sentinel-2 scene rose from 74% at 30 days to 92% at 90 and 100% at 180. A tighter window costs coverage in the tropics and buys no accuracy when the field date is only known to the year, and when plot-scale noise dwarfs seasonal change. Temporal matching matters where the forest changes between the two dates, which is the sequestration case below.

### *D. Closing the Gap (RQ2)*

**Better models.** Table V adds one information source at a time, always leave-one-region-out. These variants were run after the reveal, so they are exploratory; every variant we tried is reported. Radar alone is ahead of optical alone (54.5% vs 56.9% overall; 33.7% vs 42.0% on the 1 ha units), the two together beat either, terrain helps further, and feeding the Chloris product to the model as one more feature helps again. Of these steps, only the last one is clearly outside sampling noise (Section IV-K): the simple average of E2 and Chloris is the best blind estimator overall (45.4%, $R^2 = 0.39$), 3.7 points better than E2 with a region-bootstrap 95% interval of $-7.2$ to $-0.6$. None of these steps rescues the HCSA plots.

**Sharper inputs: canopy height and spaceborne lidar.** The largest model gain came from inputs, not algorithms. Through Earth Engine we added, for every plot, the Meta 1 m canopy height map [79] and the ETH 10 m map [80], read in and around the plot. We also added GEDI L4A footprint biomass and L2A height within 500 m and 1 km, together with the GEDI L4B grid [25], [81], and the ESA CCI Biomass map [4]. All of these are read from coordinates only. Table VI adds them to E2, always leave-one-region-out. Canopy height alone cut the overall error from 49.1% to 41.5%, and canopy height plus GEDI cut it to 40.1% ($R^2$ 0.29 to 0.53, CCC 0.58 to 0.73). The gain is outside sampling noise in every dataset: overall it is $-9.0$ points, with a region-bootstrap interval of $-13.0$ to $-6.1$. On the 1 ha units the error fell from 28.8% to 20.6%, on GEO-TREES from 42.0% to 27.7%, and on HCSA from 104.8% to 95.6%. Used as published, without calibration, the GEDI products under-estimate by 29–41% and are worse than the raw map. They help as inputs, not as answers. A height-only power law fitted on the 1 m map (AGB $= aH^b$) is no better than E2, because one height-biomass curve does not transfer across regions. The improvement carries to project scale. With no field plot, the error of a block total fell from 13% to 7% on the 1 ha units, from 20% to 9% on GEO-TREES, and from 28% to 20% on HCSA (median over blocks). The mean of 20 plots is within 3–5% of its field mean outside Indonesia.

**A few local plots.** The largest gain comes not from the model but from field data in the target region. Fig. 4(b) shows E3: with $k$ plots of the target block used for a ratio correction, the error of the block total falls from 29% (GEO-TREES) and 11% (Labrière) at $k = 0$ to 9–10% and 6% at $k = 10$–20 (median over blocks and 200 draws). On HCSA, calibration hurts until $k = 20$ (27% at $k = 0$, 53% at $k = 2$, 24% at $k = 20$): a small plot is so noisy that a handful of them is a

TABLE V
IMPROVEMENT LADDER, BLIND RRMSE (%)

| Left-grid inputs | All | GEO-T. | HCSA | Labr. |
|---|---|---|---|---|
| Optical (S2 + Landsat) | 56.9 | 51.2 | 103.0 | 42.0 |
| Radar (S1 + PALSAR) | 54.5 | 45.9 | 114.7 | 33.7 |
| Optical + radar | 52.1 | 46.4 | 103.7 | 33.9 |
| + terrain (= E2) | 49.1 | 42.0 | 104.8 | 28.8 |
| + Chloris as feature | 47.9 | 38.1 | 105.7 | 27.9 |
| + plot footprint | 47.6 | 38.6 | 103.5 | 28.1 |
| Mean of E2 and Chloris | **45.4** | **35.0** | **98.8** | 28.0 |

TABLE VI
SHARPER INPUTS: BLIND RRMSE (%), LEAVE-ONE-REGION-OUT

| Inputs | All | GEO-T. | HCSA | Labr. |
|---|---|---|---|---|
| E2 (S1/S2, Landsat, PALSAR, DEM) | 49.1 | 42.0 | 104.8 | 28.8 |
| + ESA CCI | 47.3 | 38.6 | 101.9 | 28.5 |
| + GEDI | 45.5 | 36.6 | 102.4 | 24.5 |
| + canopy height | 41.5 | 32.2 | **94.8** | 22.4 |
| + canopy height + GEDI | **40.1** | **27.7** | 95.6 | **20.6** |
| Height + GEDI only | 43.2 | 31.7 | 103.1 | 20.7 |
| Height power law (1 m) | 50.8 | 35.0 | 115.5 | 30.7 |
| GEDI L4A, as published | 59.2 | 50.0 | 107.0 | 46.6 |
| ESA CCI, as published | 49.4 | 38.8 | 105.4 | 31.5 |
| Gain of best over E2 | $-9.0$ | $-14.3$ | $-9.2$ | $-8.2$ |

Region-bootstrap 95% intervals of the gain: all $-13.0$ to $-6.1$; GEO-TREES $-27.4$ to $-7.2$; HCSA $-13.2$ to $-2.9$; Labrière $-12.8$ to $-3.1$.

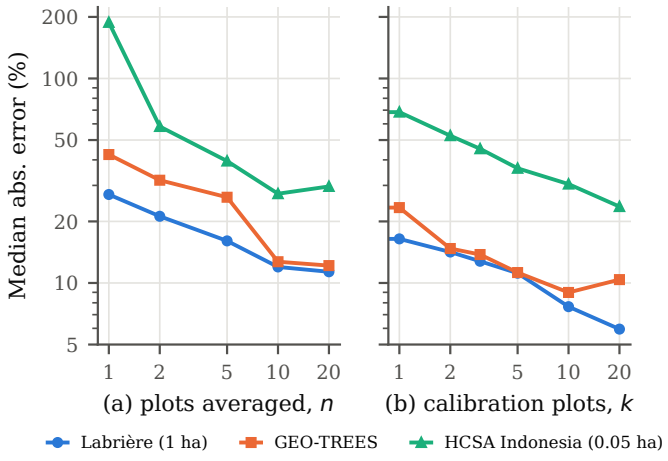



Fig. 4. Closing the gap. (a) Error of the mean of $n$ plots against its field mean; (b) error of the block total after a ratio calibration on $k$ local plots. Median over blocks and 200–300 random draws; log axes.

TABLE VII
HCSA Stems by Diameter: Count and Carbon

| Diameter | Stems ha$^{-1}$ | Stems (%) | Carbon (%) |
|---|---|---|---|
| 5–15 cm | 647 | 75.3 | 12.6 |
| 15–30 cm | 157 | 18.3 | 29.1 |
| 30–50 cm | 43 | 5.0 | 31.0 |
| $\geq$ 50 cm | 12 | 1.4 | 27.2 |

Stems per hectare averaged over the 148 plots, after the nested-area expansion of Eq. 2.

worse calibrator than none. How many local plots are needed depends on plot size as much as on the satellite.

**Aggregation.** Credits are issued for project areas, not single plots. The error of the mean of $n$ plots shrinks with $n$ until only systematic bias remains (Fig. 4(a)). For the 1 ha units the median absolute error falls from 27% for one plot to 12% for ten and 11% for twenty; for HCSA from 188% to 27–30%. The floor of roughly 10% outside Indonesia and 25–30% in Indonesia is the regional bias that only local calibration removes.

**Intervals** are analysed together with clouds in Section IV-H.

### *E. Counting Trees from Space (RQ2)*

**What the field counts.** The median HCSA plot holds 700 stems per hectare (200 of 15 cm or more) of five species. Table VII shows where its carbon is: stems of 5–15 cm are three quarters of the count but an eighth of the carbon, and the 6% of stems of 30 cm or more hold 58%. A method that finds the large trees finds most of the carbon; a method that counts stems is mostly counting saplings.

**The wood.** Table VIII groups every HCSA stem by the wood density of its species into light ($\rho < 0.5\,\mathrm{g\,cm^{-3}}$), medium (0.5–0.7), and heavy ($\geq$ 0.7) wood. Light wood, mostly Lauraceae, Moraceae, and Anacardiaceae, is a quarter of the stems but a sixth of the carbon. Heavy wood, led by Dipterocarpaceae, Lauraceae, and Myrtaceae, is a third of the stems and two fifths of the carbon. Dipterocarpaceae appear

TABLE VIII
HCSA Stems by Wood Density Class

| Class (g cm$^{-3}$) | Spp. | Stems | C | $\bar{\rho}$ | Leading families (by carbon) |
|---|---|---|---|---|---|
| light (<0.5) | 83 | 24 | 16 | 0.46 | Lauraceae, Moraceae, Anacardiaceae |
| medium (0.5–0.7) | 179 | 44 | 44 | 0.60 | Dipterocarpaceae, Moraceae, Anacardiaceae |
| heavy (≥0.7) | 127 | 32 | 40 | 0.78 | Dipterocarpaceae, Lauraceae, Myrtaceae |

Stems and C: share of stems per hectare and of carbon, %. One stem of *Myristica argentea* (plot PB01) is recorded at $\rho = 2.47$, which is physically impossible; it holds 1.1% of its plot and is left as published.

in both the medium and the heavy class because the family spans both. The stem-weighted mean is 0.627 g cm$^{-3}$, against the 0.574 that E4 must assume. Giving every stem that default instead of its species value leaves 91% of the total carbon and 78–115% of a single plot's (10th to 90th percentile). Wood density alone, the one input a satellite cannot see at all, is therefore worth about a tenth of the carbon in total and a fifth in a single plot.

**What the satellite counts.** On the HCSA plots E4 found a median of 20 trees per hectare: 2.7% of the stems, 11% of those of 15 cm or more, with a rank correlation of 0.18 against the field count. The 0.05 ha HCSA footprint holds few crowns: 60 of the 148 plots received no tree and 42 received one. The Meta map is a model output stored in whole metres and is smoother than a crown. Without smoothing, every stair step of the integer surface becomes a top and the count rises to 270 per hectare. A tree count from this map is therefore a processing choice, not a measurement, and no setting can be checked without stem lists.

**The carbon E4 finds.** Table IX reports amounts, not errors. E4 found 25% of the field carbon overall, and 23–26% in each dataset; it under-counts trees and under-sizes them in the same proportion everywhere. Its log correlation with the field value was 0.83 on GEO-TREES and 0.80 on Labrière, so it ranks large plots well. Scaled by the factor learned on other regions (median $\times$3.8), it found 97% of the total, but only 36% of plots fell within $\pm$25% of the field value, against 60% for the map, and on HCSA 9% against 22%. Counting trees adds an interpretable middle step, but it is a worse estimator than learning biomass from the pixels directly.

**Why greenness is not enough.** Of the 136 plots with a clear Sentinel-2 view in the matching window, the median NDVI (Eq. 7) was 0.77 below 100 Mg ha$^{-1}$, 0.79 at 100–200, and 0.83 at 200–300 and above 300. Over all plots its rank correlation with AGB was 0.26, and above 150 Mg ha$^{-1}$ it was $-0.13$. A closed canopy is equally green whatever the biomass beneath it, which is why the models lean on radar, canopy height, and GEDI.

### *F. Ten Simulated Projects*

The tables above score plots. A buyer of credits sees a project. We therefore simulate 1,000 projects of 1,000 ha. Each simulation draws one of the 16 regions with at least 10 plots, each region equally likely, and 10 of its plots at random; the 10 plots stand for the project. For those plots it reports the carbon of the whole project by the field (right grid), the map (E2 with

TABLE IX
CARBON FOUND BY E4 AND BY THE MAP, AGAINST THE FIELD ($t\,CO_2e\,ha^{-1}$ AND %)

| Subset ($n$) | Field | Estimate | Sat. | Total | Med. | ±25% |
|---|---|---|---|---|---|---|
| All (475) | 449 | E4, as is | 111 | 25 | 23 | 1 |
| | | E4, unsmoothed | 178 | 41 | 37 | 4 |
| | | E4, scaled | 427 | 97 | 89 | 36 |
| | | Map (E2 + CH + GEDI) | 417 | 98 | 100 | 60 |
| GEO-TREES (142) | 473 | E4, as is | 115 | 24 | 24 | 0 |
| | | E4, unsmoothed | 180 | 38 | 35 | 2 |
| | | E4, scaled | 435 | 92 | 89 | 42 |
| | | Map (E2 + CH + GEDI) | 467 | 99 | 99 | 72 |
| HCSA (148) | 277 | E4, as is | 64 | 23 | 7 | 3 |
| | | E4, unsmoothed | 98 | 35 | 20 | 10 |
| | | E4, scaled | 245 | 88 | 26 | 9 |
| | | Map (E2 + CH + GEDI) | 199 | 72 | 90 | 22 |
| Labrière (185) | 567 | E4, as is | 147 | 26 | 26 | 0 |
| | | E4, unsmoothed | 241 | 43 | 43 | 1 |
| | | E4, scaled | 567 | 100 | 103 | 52 |
| | | Map (E2 + CH + GEDI) | 554 | 98 | 101 | 81 |

Field and Sat.: mean $t\,CO_2e\,ha^{-1}$. Total: satellite carbon as % of field carbon over the plot footprints. Med.: median plot %. ±25%: % of plots within 25% of the field value. CH: canopy height.

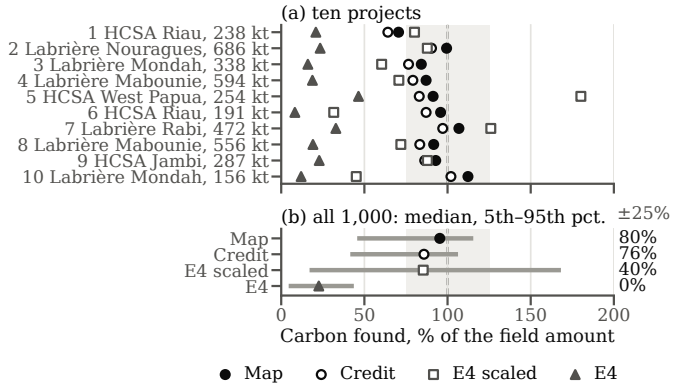



Fig. 5. Carbon found in simulated 1,000 ha projects of 10 random plots, as % of the field amount: (a) the ten projects of Table X, with their field carbon in kt $CO_2$e; (b) all 1,000, median and 5th–95th percentile, with the share of projects within ±25% at right. The shaded band is ±25% of the field.

canopy height and GEDI), E4 as is, E4 scaled, and conformal crediting on the map at $\delta = 0.1$ (Section IV-J), as tonnes of $CO_2$e and as a percentage of the field amount. Table X lists the first ten simulations, Fig. 5 draws them beside the spread of all 1,000, and Appendix A lists every one of their plots.

Across all 1,000 simulations the map found a median 95% of the field carbon, within ±10% in half of the projects and within ±25% in 80%. E4 never came within 25%. Conformal crediting issued a median 86% of the field amount and exceeded it in 12% of projects, near its 10% target; these random 10-plot projects are not the 31 calibration projects of Section IV-J, so the guarantee holds only approximately. The raw map exceeded the field amount in 38% of projects, which is the over-crediting a platform without a bound would issue. Inside a project the plots disagree far more than the totals: in simulation 1 the map gave one Riau plot six times its field

TABLE X
TEN SIMULATED 1,000 HA PROJECTS OF 10 RANDOM PLOTS: CARBON FOUND BY EACH GRID, % OF THE FIELD AMOUNT

| # | Region | Field, kt | Map | E4 | E4 sc. | Credit |
|---|---|---|---|---|---|---|
| 1 | HCSA Riau | 238 | 71 | 21 | 80 | 64 |
| 2 | Labrière Nouragues | 686 | 99 | 23 | 88 | 90 |
| 3 | Labrière Mondah | 338 | 84 | 16 | 60 | 77 |
| 4 | Labrière Mabounie | 594 | 87 | 19 | 71 | 79 |
| 5 | HCSA West Papua | 254 | 91 | 46 | 180 | 83 |
| 6 | HCSA Riau | 191 | 96 | 8 | 32 | 87 |
| 7 | Labrière Rabi | 472 | 107 | 33 | 126 | 97 |
| 8 | Labrière Mabounie | 556 | 92 | 19 | 72 | 83 |
| 9 | HCSA Jambi | 287 | 93 | 23 | 88 | 86 |
| 10 | Labrière Mondah | 156 | 112 | 12 | 45 | 102 |
| 1,000 projects: median | | | 95 | 23 | 85 | 86 |
| 5th–95th percentile | | | 46–115 | 4–44 | 17–168 | 42–106 |
| within ±10% of the field | | | 52 | 0 | 17 | 33 |
| within ±25% of the field | | | 80 | 0 | 40 | 76 |
| above the field | | | 38 | 0 | 37 | 12 |

Field: kt $CO_2$e from the measured trees, Eq. 1, $\times 0.47 \times \frac{44}{12}$. Map: E2 + canopy height + GEDI. E4 sc.: E4 times the field-to-E4 ratio of the other regions (about 3.8). Credit: conformal crediting on the map, $\delta = 0.1$. Seeds 1–10; none drew a GEO-TREES region.

value and another half, and the project still came out at 71%.

### G. Toward Production: Fixes and an Unseen Test

The results so far name five limits: the error of small Indonesian plots, a regional bias the map cannot see, a tree count that finds a quarter of the carbon, a credit that exceeded the field amount in 12% of random projects, and a platform height formula that was never checked. We attacked each after the reveal, so every result in this subsection is exploratory except the last: the method was then frozen and tested blind on 35 plots that no model had seen (Section IV-G1).

**New inputs.** Two open layers appeared after our extraction. The AlphaEarth Foundations satellite embedding [92] gives every 10 m pixel and year since 2017 a 64-dimensional vector learnt from Sentinel-1/2, Landsat, GEDI, and other sources; we average it in the plot and within 100 m, in the year nearest the field date (2017 for older plots). The Meta/WRI canopy height map v2 (CHMv2) [93] replaces the 2024 map: its makers report a mean absolute error of 3.0 m against 4.3 m and no mean bias against airborne lidar, and on our plots it reads a mean of 20.1 m against 16.2 m. It is not in Earth Engine, so we read its open cloud-optimised tiles onto the same 1 m grid. Table XI gives the leave-one-region-out result. The embeddings with canopy height and GEDI cut the plot error from 40.1 to 35.8% (region bootstrap $-8.0$ to $-1.7$ points) and lifted projects within ±25% of the field from 80 to 84%, and HCSA projects from 44 to 56%, from Earth Engine layers alone. CHMv2 added about a point at plot level and nothing at project level. A ridge regression on the same inputs, which an app could evaluate as a band expression, was worse than the original map, so production needs the tree model.

**Published height equations, no training.** An app without training data could convert canopy height to carbon with a published lidar equation. Table XII applies four unchanged to the mean canopy height TCH, the share of pixels above

TABLE XI
NEW INPUTS, LEAVE-ONE-REGION-OUT: PLOT RRMSE (%) AND THE 1,000 SIMULATED PROJECTS

| Inputs | Plot rRMSE All | GEO | HCSA | 1 ha | Projects Med. | ±25 | HCSA |
|---|---|---|---|---|---|---|---|
| Map of Table VI | 40.1 | 27.7 | 95.6 | 20.6 | 95 | 80.0 | 44.1 |
| + embeddings | 36.2 | 25.1 | 86.9 | 18.1 | 93 | 80.7 | 46.1 |
| Embeddings only | 37.0 | 25.7 | 88.6 | 18.5 | 96 | 83.7 | 56.4 |
| Emb. + CH + GEDI | 35.8 | 25.9 | 85.2 | 17.6 | 95 | 84.2 | 55.9 |
| Emb. + CHMv2 + GEDI | 35.1 | 24.6 | 84.4 | 17.1 | 95 | 83.9 | 55.0 |
| Emb. + CHMv1/2 + GEDI | 34.7 | 24.9 | 82.7 | 17.2 | 95 | 85.2 | 58.7 |
| Ridge, emb. + CH + GEDI | 44.6 | 41.2 | 86.4 | 29.3 | 101 | 59.2 | 53.6 |

Projects: median % of the field carbon, and the share of projects (all; HCSA only) within ±25%. CH: Meta 2024 and ETH canopy height. Emb. + CH + GEDI uses Earth Engine layers only and is the production map.

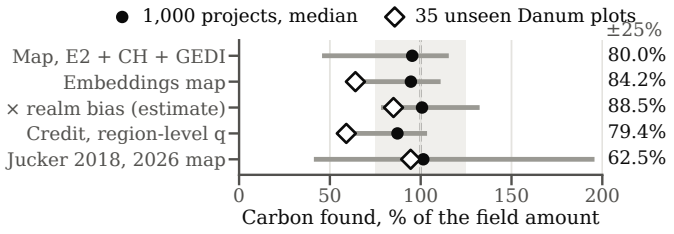



Fig. 6. Each fix of Table XIII on the 1,000 projects (median and 5th–95th percentile; right: share within ±25% of the field) and its total on the 35 unseen Danum plots (Table XIV). The shaded band is ±25% of the field. Jucker's equation is unfitted and was run on Danum after the reveal.

20 m, and the share below 19 m in each plot: Jucker *et al.*'s Sabah model ACD $= 0.567\,\mathrm{TCH}^{0.554}\mathrm{BA}^{1.081}\mathrm{WD}^{0.186}$ with WD $= 0.385\,\mathrm{TCH}^{0.097}$ and basal area from height alone, BA $= 1.112\,\mathrm{TCH}$ (their Eq. 9), or from height and cover, BA $= 1.287\,\mathrm{TCH}^{0.987}(1 + 1.983\,c)$ with $c$ the cover above 20 m less its expectation $1/(1 + e^{12.431}\mathrm{TCH}^{-4.061})$ (Eq. 10) [94]; Coomes *et al.*'s Sepilok model ACD $= 25.93\,\mathrm{GF}_{19}^{-0.209}\mathrm{TCH}^{0.437}$ [95]; and our platform's formula. All published ones output carbon density, so AGB is ACD$/0.47$. On the 2024 map every equation fell short or scattered; on CHMv2, which reads taller canopies, Jucker's Eq. 10 found 96% of the field total and 94% on the unseen Danum plots, its home forest. Unfitted equations still scatter at project scale (5th–95th percentile 41–196% for Jucker's Eq. 10), so they suit a first estimate, not a credit.

Our platform's formula, AGB $= 3.836\,H^{1.131}$, cites Asner and Mascaro [27] but matches no equation there: their general model is ACD $= 3.836\,\mathrm{TCH}^{0.281}\mathrm{BA}^{0.972}\mathrm{WD}^{1.376}$ in $\mathrm{Mg\,C\,ha^{-1}}$, with basal area and wood density, as quoted by Jucker *et al.* [94]. The platform's curve behaves like a carbon model but is treated as biomass and multiplied by 0.47 again, so it found 35% of the field carbon; read as carbon it would find 75% on the 2024 map and 95% on CHMv2. Its input is also broken: it reads a band `MU` from the GEDI L4A monthly collection, which has no such band (`MU` belongs to the 1 km L4B grid), so every forest estimate stops with an error.

**Counting trees on the new map.** The tree count of Section IV-E improves on CHMv2: a median of 39 trees per hectare instead of 20, 62% of the field carbon instead of 25% (61% GEO-TREES, 48% HCSA, 68% Labrière), and 24 plots with no tree instead of 66. The remaining shortfall, a factor of 1.6, approaches what tree-centric airborne lidar leaves, a factor of 1.07–1.32 [95]; the understorey is still invisible.

**Regional bias and local plots.** The map's error is not random within a region: Indonesian projects are under-estimated as a group. Multiplying the map by $e^{-\mu}$, with $\mu$ the mean log ratio of map to field over the other regions, removes the average bias; taking $\mu$ from the other regions of the same biogeographic realm (a project in Riau borrows the bias of the other Indonesian provinces) is better, lifting projects within ±25% from 84.2 to 88.5% and HCSA projects from 56 to 68% (Table XIII). Local field plots help less than expected. A ratio correction on 3 or 10 random plots outside the project removes the bias but adds the plots' own noise, and on 0.05 ha HCSA plots the noise wins (32 and 51% within ±25%). Shrinking the local log ratio $z_k$ toward the realm prior, $z^* = (\mu/\tau^2 + z_k/v)/(1/\tau^2 + 1/v)$ with $\tau^2$ the spread of regional biases and $v = s^2/k$ the plot noise of the same plot protocol, recovers the realm-corrected result but does not beat it. Plots pay off when they are large: ten 1 ha plots narrowed Labrière projects from 86–113 to 91–113% of the field.

**A credit that holds its target.** The project-level conformal credit of Table X exceeded the field amount in 12% of random projects, and 13.6% on the embeddings map. Its calibration units were sub-projects whose errors share their region's bias, so they are not exchangeable with a new region. Taking one score per region with at least 10 plots, $z_r = \ln(\sum \hat{Y} / \sum Y)$ over the region, and the conformal quantile of the other regions' scores in Eq. 16, over-credited 9.7% of projects at a median 87% of the field carbon (Table XIII). Applying the same rule to the realm-corrected map gave a higher credit for HCSA but over-credited 27.9% of HCSA projects, because the correction is estimated from as few as four regions; we keep the uncorrected map for the credit.

*1) Unseen Test: Danum, Sabah:* Fig. 6 sets each fix beside its result on unseen plots. To test the method rather than our tuning of it, we froze it before opening any new data: the production map trained on every plot except the three GEO-TREES plots of the Danum valley, the realm correction $\mu$ from the six Indonesian provinces ($\mu = -0.283$, a factor of 1.33), and the region-level credit ($q = 0.081$, a factor of 0.92), both from the leave-one-region-out predictions of regions with at least 10 plots. The test data are the 35 "Danum Revised" plots [96]: circles of 30 m radius in the Danum Valley, Sabah, measured from January to June 2020, with diameter, height, and species for every stem of 10 cm or more. The right grid applied Eq. 1 to the 4,201 live stems inside the radius with their measured heights and wood density from the Global Wood Density Database (species, genus, family, plot mean), and was sealed in a separate ledger before any feature was read. The left grid read only the manifest. The prediction file was committed, then the seal opened; the ledger verifies seal,

TABLE XII
PUBLISHED CANOPY-HEIGHT EQUATIONS, UNFITTED, ON THE 2024 MAP (CHMV1) AND CHMV2: % OF THE FIELD CARBON

| Equation | Output | CHMv1 rRMSE | CHMv1 Total | CHMv1 Proj. ±25% | CHMv2 rRMSE | CHMv2 Total | CHMv2 Proj. ±25% | Danum total |
|---|---|---|---|---|---|---|---|---|
| Jucker 2018 Eq. 9, BA from height | $\mathrm{Mg\,C\,ha^{-1}}$ | 68.8 | 48.6 | 9.6 | 52.9 | 68.6 | 42.9 | 70 |
| Jucker 2018 Eq. 10, BA from height and cover | $\mathrm{Mg\,C\,ha^{-1}}$ | 61.3 | 65.7 | 24.1 | 51.3 | 95.6 | 62.5 | 94 |
| Coomes 2017, gap fraction | $\mathrm{Mg\,C\,ha^{-1}}$ | 52.9 | 99.4 | 64.7 | 57.2 | 121.1 | 48.2 | 114 |
| Platform, as coded: $3.836\,H^{1.131}$ | "AGB" | 81.4 | 35.1 | 5.1 | 72.2 | 44.6 | 9.9 | 42 |
| Platform, read as carbon | $\mathrm{Mg\,C\,ha^{-1}}$ | 51.1 | 74.6 | 35.9 | 42.1 | 95.0 | 73.0 | 89 |
| Trained map (Table VI) | | 40.1 | 93.1 | 80.0 | | | | |

Total: sum of estimates over the sum of field values, 475 plots. Proj. ±25%: share of the 1,000 simulated projects within ±25% of the field carbon. $\mathrm{GF}_{19}$: share of pixels below 19 m, floored at 0.01. Danum: the 35 unseen plots of Section IV-G1, computed after the reveal.

TABLE XIII
BIAS CORRECTION, LOCAL PLOTS, AND CREDITS ON THE PRODUCTION MAP: 1,000 SIMULATED PROJECTS, % OF THE FIELD CARBON

| Estimate | All projects Median | 5–95th | Within ±25% | Above field | HCSA projects Median | Within ±25% | Above field |
|---|---|---|---|---|---|---|---|
| Production map (embeddings + CH + GEDI) | 95 | 61–111 | 84.2 | 32.5 | 78 | 55.9 | 12.3 |
| $\times\, e^{-\mu}$, $\mu$ from all other regions | 106 | 67–126 | 85.6 | 67.7 | 87 | 71.0 | 28.5 |
| $\times\, e^{-\mu}$, $\mu$ from the same realm (**estimate**) | 101 | 78–132 | **88.5** | 52.2 | 104 | 67.9 | 56.4 |
| 3 local plots, ratio | 98 | 49–194 | 66.9 | 46.5 | 96 | 32.1 | 48.0 |
| 3 local plots, shrunk to the realm prior | 101 | 75–135 | 86.7 | 53.3 | 103 | 67.6 | 55.0 |
| 10 local plots, ratio | 100 | 60–159 | 78.9 | 51.2 | 102 | 51.5 | 52.3 |
| 10 local plots, shrunk to the realm prior | 102 | 68–134 | 85.6 | 54.9 | 97 | 65.4 | 46.8 |
| Credit, project-level $q$ (as Table X) | 88 | 57–104 | 80.7 | 13.6 | 73 | 46.1 | 5.6 |
| Credit, region-level $q$ (**credit**) | 87 | 56–104 | 79.4 | **9.7** | 72 | 42.7 | 5.0 |
| Credit, region-level $q$ on the realm-corrected map | 85 | 68–111 | 85.1 | 10.9 | 88 | 72.1 | 27.9 |

Same 1,000 projects as Table X; all maps leave-one-region-out. Local plots are drawn from the project's region outside its 10 plots, so only projects in regions with enough remaining plots count (869 for $k = 3$, 568 for $k = 10$). Credits target $\delta = 0.1$.

TABLE XIV
UNSEEN TEST: 35 DANUM PLOTS, FROZEN METHOD, % OF THE FIELD CARBON

| Estimate | rRMSE | 1,000 draws of 10 plots Total | Median | 5–95th | ±25% |
|---|---|---|---|---|---|
| Map | 52.9 | 64.1 | 64 | 55–77 | 7.6 |
| Estimate, $\times\, e^{-\mu}$ | 40.1 | **85.0** | 85 | 74–103 | **92.8** |
| Credit, $\times\, e^{-q}$ | 56.8 | 59.0 | 59 | 51–71 | 2.0 |

Field: mean 366, median 339 $\mathrm{Mg\,ha^{-1}}$. No draw of the credit exceeded the field amount; 7.0% of the estimate's draws did.

commit, reveal.

Danum is tall dipterocarp forest (mean 366 $\mathrm{Mg\,ha^{-1}}$, CHMv2 mean height 25.9 m), where the map saturates: alone it found 64% of the field carbon (Table XIV). The realm correction learnt in Indonesia raised this to 85%, and 93% of random 10-plot projects fell within ±25% of the field amount. The credit, 59%, never exceeded the field. Danum is one valley, so this is one test of transfer within the realm, not of the method everywhere; it does show that a bias learnt in Sumatra, Kalimantan, and Papua carried to Sabah, and that the credit erred on the safe side.

**The production method.** What we would deploy follows from these results:

1) *Features*, from Earth Engine only, per 1 ha cell of the polygon: the AlphaEarth embedding of the year, Meta and ETH canopy height statistics, and GEDI L4A footprints within 500 m and 1 km with the L4B grid.
2) *Map*: LightGBM on lnAGB trained on all field plots, with smearing back-transformation; the polygon's stock is the sum over cells.
3) *Estimate*: the map times $e^{-\mu}$, $\mu$ the mean regional bias of the polygon's realm from leave-one-region-out predictions.
4) *Credit*: the map times $e^{-q}$, $q$ the region-level conformal quantile at $\delta = 0.1$, recalibrated whenever field-verified regions are added.
5) *Growth*: a regional default, since no satellite estimate beat "no change" (Section IV-I); the IPCC 2019 Refinement gives 0.7, 2.7, and 3.4 t d.m. $\mathrm{ha^{-1}\,yr^{-1}}$ for primary, secondary older than 20 years, and younger secondary Asian tropical rainforest [66], with satellites used to detect loss.
6) *Field plots*, where a project adds them, of 0.25–1 ha, combined by shrinkage toward the realm prior.

### H. Clouds and Confidence (RQ3)

**How often is a plot clear?** Table XV gives the plot-level climatology. On a median Indonesian HCSA location, only 21.5% of Sentinel-2 acquisitions see the plot completely clear (Sabah 26%, Gabon and Congo 17%, the Americas 39%). The year is not flat (Fig. 7): in Indonesia the clear share falls to 7.6% in January and 14–15% in December and February, and peaks at 33.5% in May, a factor of 4.4. Revisit compensates

TABLE XV
PLOT-LEVEL SENTINEL-2 CLEAR-SKY CLIMATOLOGY, 2021–2023 (MEDIAN OVER LOCATIONS)

| | | | Gap (d) | | P(clear in $\pm W$) | |
|---|---|---|---|---|---|---|
| **Region** | **Loc.** | $p$ | **Med.** | **P90** | $\pm 15$ d | $\pm 30$ d |
| Indonesia (HCSA) | 24 | 0.215 | 10 | 40 | 0.81 | 0.96 |
| Sabah (SAFE) | 10 | 0.262 | 15 | 41 | 0.84 | 0.99 |
| Africa | 22 | 0.166 | 18 | 62 | 0.61 | 0.79 |
| Americas | 40 | 0.394 | 10 | 25 | 0.82 | 0.94 |

$p$: share of acquisitions with every plot pixel clear. Independence model at $\pm 30$ d: Indonesia 0.98, Sabah 0.98, Africa 0.89, Americas 1.00.

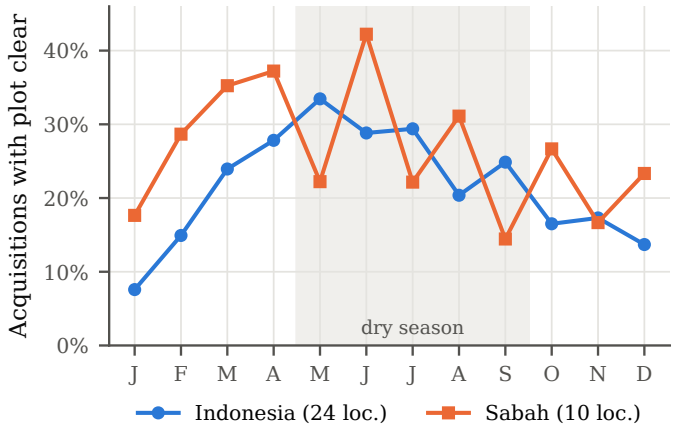



Fig. 7. Share of Sentinel-2 acquisitions in which the whole plot is clear, by month, 2021–2023.

for cloudiness. With two Sentinel-2 satellites, the median gap between clear looks is 10 days and the 90th percentile 40 days, so 81% of random dates have a clear look within $\pm 15$ days and 96% within $\pm 30$. Clouds cluster in time: the measured probability falls below the independence model in every region, most in Africa ($\pm 30$ days: 79% against 89%).

**The scene filter is the real problem.** Scene-level cloud percentage agrees only loosely with the plot (Spearman $-0.67$). In Indonesia, the rule `CLOUDY_PIXEL_PERCENTAGE` $< 30$ keeps 11.5% of acquisitions, but 34% of the scenes it keeps are cloudy over the plot, and it throws away 65% of the acquisitions in which the plot is clear. At $< 40$, the threshold our own platform uses for segmentation, 40% of the kept scenes are cloudy over the plot. Masking at the plot instead of filtering scenes gives 2.8 times as many usable clear looks (21.5% against 7.6% of acquisitions) and removes the cloudy ones that pass the filter.

**What clouds cost the estimate.** Table XVI removes the optical block from a growing share of target plots. Because the model leans on radar, terrain, and the annual product, losing every optical feature raises the blind error from 48.0% to 50.0% rRMSE. Modality dropout and the cloud-aware switch both handle partial cloud well; the switch is best on the cloud-hit plots at low $q$ (37.7% against 43.3% for the naive model at $q = 0$, on the 23 plots that really had no clear scene). Imputing

TABLE XVI
BLIND RRMSE (%) WHEN CLOUDS REMOVE THE OPTICAL BLOCK FROM A SHARE $q$ OF TARGET PLOTS

| $q$ | **Naive** | **Dropout** | **Switch** | **Impute** |
|---|---|---|---|---|
| 0 | 48.0 | 48.0 | **47.7** | 51.5 |
| 0.25 | 48.5 | 48.6 | **47.9** | 61.2 |
| 0.5 | 49.6 | 49.1 | **49.1** | 67.9 |
| 0.75 | 49.3 | 49.4 | **49.1** | 77.5 |
| 1 | **50.0** | **50.0** | 50.4 | 84.6 |

TABLE XVII
90% INTERVAL METHODS, LEAVE-ONE-REGION-OUT (COVERAGE; MEDIAN UPPER/LOWER RATIO; WINKLER SCORE)

| **Method** | **All** | **GEO-T.** | **HCSA** | **Labr.** | **Ratio** | **Winkler** |
|---|---|---|---|---|---|---|
| Split | 0.884 | 0.958 | 0.682 | 0.989 | 16.3 | 4.21 |
| Mondrian | 0.882 | 0.958 | 0.676 | 0.989 | 16.3 | 4.42 |
| Normalized | 0.794 | 0.782 | 0.723 | 0.859 | 3.3 | 3.53 |
| CQR | **0.899** | 0.937 | **0.811** | 0.941 | 5.3 | **3.05** |

optical features from radar is harmful: the bias grows to $-61\%$ at full cloud, a warning for SAR-to-optical gap filling used as a model input.

**Which intervals are honest?** Table XVII compares the four interval methods. Split conformal, the method of the confirmatory run, reaches 88% coverage but is wide (upper bound 16 times the lower) and fails on HCSA (68%). Normalized conformal is narrow but under-covers (79%), because a difficulty model fitted on other regions does not transfer. CQR is the only method that reaches the target overall (89.9%), with intervals three times narrower than split (ratio 5.3) and the best Winkler score. Mondrian calibration by optical availability adds nothing: only 23 plots lack optical data, too few to calibrate a group. HCSA remains under-covered by every method (68–81%): its plots are noisier than any training block, a shift that marginal conformal guarantees do not cover.

**Intervals under cloud.** Fig. 8 shows the effect. When every target plot is clouded, plain CQR, calibrated on plots as observed, falls to 86.5% coverage (HCSA 70%). The cloud-aware CQR, calibrated on synthetic cloud, keeps 90.5% (HCSA 77%, GEO-TREES 95%, Labrière 98%). It does this by widening the interval honestly, to a ratio of 7.8 against 5.4. On plots as observed, it covers 90.5% at ratio 5.7. A cloudy plot gets a wider interval, which is what an honest confidence statement should do.

**A metadata score is not a confidence.** Over all plots, the rebuilt platform-style confidence score correlates with the actual error in the *wrong* direction (Spearman $+0.39$): plots scored as most reliable were the least accurate. That figure is confounded. The recent, cloud-checked HCSA plots score high and are the hardest to predict. Once predicted AGB and the dataset (or the region) are removed, the partial rank correlation is $-0.03$ to $+0.07$, so the score carries no information about error beyond which dataset a plot belongs to. The learned difficulty $\hat{\sigma}(x)$ fares no better at plot level (partial correlation $+0.02$ to $+0.07$). Neither a metadata score nor a learned

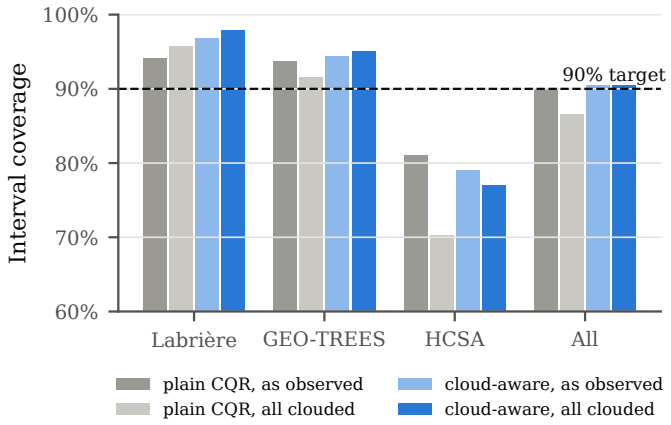



Fig. 8. Coverage of 90% intervals on target plots as observed and with every optical feature removed. Cloud-aware calibration keeps the target overall; plain CQR falls to 86.5%.

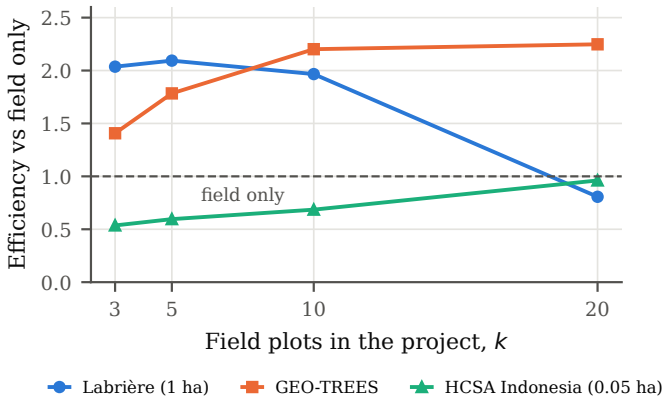



Fig. 9. Efficiency of the ratio estimator (satellite + $k$ field plots) relative to the field-only mean, by empirical MSE over 500 draws. Above 1, the satellite saves field plots.

difficulty can rank individual plots by error here. Confidence therefore has to be stated as a calibrated interval and checked against field data, as found for global biomass maps [14], [97].

**Project-level confidence and over-crediting.** For 19 projects (blocks with at least 8 plots), the blind satellite mean missed the field mean by a median of 23%. The naive interval, with plot errors assumed independent, had a median half-width of 28% but covered only 74% of projects (HCSA 33%), and two projects would have been over-credited even after the VM0047 deduction. The two-component interval of Eq. 14 covered all 19, at an honest half-width of 71–86%, which under VM0047 means 61–76% of the credits are deducted. This agrees with earlier coverage studies of model-based estimators that omit residual spatial correlation or model bias [54], [55]; what our test adds is the size of the effect when a model is transferred across regions. A satellite-only project estimate is therefore either over-confident or almost uncreditable.

Field plots change this (Fig. 9, Table XVIII). With the satellite as auxiliary data, the ratio estimator is 1.4 to 2.3 times as efficient as the field-only mean on GEO-TREES ($k = 3$ to 20) and about 2 times on Labrière ($k = 3$ to 10); at $k = 20$ the small Labrière regions are nearly fully sampled and the advantage disappears. Equivalently, the satellite replaces roughly half of the field plots needed for a given precision: summed over regions, the plots needed for ±10% at 90% fall from 408 to 216 (GEO-TREES) and from 381 to 162 (Labrière). On HCSA, where the satellite cannot rank small plots (within-region correlation 0.2), the ratio estimator is less efficient than field-only at every $k$ (0.54–0.96), only GREG gains at $k = 20$ (1.25), and the plots needed rise from 1,483 to 1,877. The gain grows with plot size, as found for airborne lidar and InSAR in Tanzania [53]. GREG under-states its own variance at small $k$ (coverage 66–73% at $k = 3$), so the ratio estimator is the safer choice. Finally, the VM0047 deduction does not make an estimate conservative: with $k = 10$ plots,

TABLE XVIII
SATELLITE AS AUXILIARY DATA: COVERAGE OF 90% CI AND EFFICIENCY VS FIELD-ONLY (500 DRAWS)

| | | Field only | | Ratio (satellite + $k$) | | |
|---|---|---|---|---|---|---|
| **Dataset** | $k$ | **Cov.** | **RMSE** | **Cov.** | **RMSE** | **Eff.** |
| GEO-TREES | 5 | 0.91 | 16.3 | 0.90 | 12.2 | 1.78 |
| | 10 | 0.88 | 10.4 | 0.89 | 7.0 | 2.20 |
| | 20 | 0.90 | 5.7 | 0.90 | 3.8 | 2.25 |
| HCSA | 5 | 0.85 | 36.7 | 0.80 | 47.6 | 0.60 |
| | 10 | 0.86 | 22.4 | 0.86 | 27.1 | 0.69 |
| | 20 | 0.89 | 11.1 | 0.87 | 11.3 | 0.96 |
| Labrière | 5 | 0.89 | 17.9 | 0.86 | 12.3 | 2.09 |
| | 10 | 0.88 | 10.1 | 0.88 | 7.2 | 1.97 |
| | 20 | 0.88 | 3.1 | 0.89 | 3.5 | 0.81 |

RMSE in % of the project mean. Eff.: MSE(field)/MSE(ratio); $> 1$ saves plots.

the deducted estimate still exceeded the true stock in 32–43% of draws on the large-plot datasets, for all three estimators, because the deduction only starts above 10% uncertainty. Analytically, the deducted estimate is $\hat{C}(1 - U + 0.10)$, ten points of $\hat{C}$ above the one-sided 95% lower bound, so its exceedance probability must lie between about 5% and 50%; the simulation places it for realistic designs. Critiques of registry deductions so far have been qualitative [86], [98], and over-crediting has been documented for other rules [99].

### *I. Sequestration (RQ4)*

**A right-grid failure first.** After the reveal, the 27 oil-palm plots showed a mean field stock of 505 Mg ha$^{-1}$, more than old-growth forest. Every stem in those plots is an oil palm whose recorded "diameter" (median 76 cm) includes the frond bases, and Chave Eq. 4 is a model for dicotyledonous trees. The standard formula is therefore not valid there. The evaluation script had fixed an "excluding oil palm" subset before the reveal, and we report it as the primary result: 196 plots in 15 blocks, mean field $\Delta$AGB $= 1.87$ Mg ha$^{-1}$ yr$^{-1}$ (0.88 t C ha$^{-1}$ yr$^{-1}$), median census gap 8.4 years.

**Blind satellite change estimates do not beat "no change".** Table XIX gives the scores. At plot level the Chloris difference

TABLE XIX
BLIND SEQUESTRATION, SAFE EXCLUDING OIL PALM (MG AGB $\text{HA}^{-1}$ $\text{YR}^{-1}$; FIELD MEAN 1.87)

| | **Plot** (196) | | | **Block mean** (15) | |
|---|---|---|---|---|---|
| **Estimator** | **RMSE** | **Bias** | $r$ | **MAE** | **Bias** |
| No change | 8.12 | −1.87 | – | 2.05 | −1.92 |
| S0 Chloris difference | 8.06 | −1.01 | 0.07 | 1.61 | −1.20 |
| S2 E2 stock difference | 16.36 | −5.90 | −0.02 | 7.40 | −7.08 |
| S2c consistent sensors† | 13.96 | −3.50 | 0.09 | 4.48 | −3.77 |
| S3 change model, LOBO† | 8.39 | −0.50 | −0.07 | 1.33 | −0.27 |
| Regional default rate† | **7.91** | **0.00** | – | **0.98** | −0.05 |

†Exploratory, defined after the reveal.

(S0) has almost the same error as predicting zero (RMSE 8.06 vs 8.12 $\text{Mg ha}^{-1}\,\text{yr}^{-1}$, $r = 0.07$). The E2 stock difference (S2) is twice as bad and has the wrong sign: it predicts a loss of 4 $\text{Mg ha}^{-1}\,\text{yr}^{-1}$ where the forest gained 1.9. The cause is a sensor artefact: Sentinel-1 existed for only 13% of the plots at $t_0$ but for all of them at $t_1$, and no L-band mosaic before 2015 is available in the catalogue, so the two stock predictions are made from different inputs. Restricting the stock model to sensors present at both dates (S2c: Landsat and terrain) halves the bias but not the noise.

**Why.** The blind E2 stock error on these plots has a standard deviation of 146 $\text{Mg ha}^{-1}$. If the errors at $t_0$ and $t_1$ were independent, differencing them over 8.4 years would add noise of about 25 $\text{Mg ha}^{-1}\,\text{yr}^{-1}$, thirteen times the signal. The errors are partly correlated, because the same place is seen twice, which is why the observed S2c error (14 $\text{Mg ha}^{-1}\,\text{yr}^{-1}$) is lower, but it is still seven times the signal. A stock estimator with plot-scale error of 30–100% cannot resolve growth of 1% per year.

**What does work is not a satellite.** Two estimators trained on the other SAFE blocks' measured change (leave-one-block-out, run after the reveal and therefore exploratory) were added: a LightGBM on Landsat and Chloris change features (S3), and the plain mean rate of the other blocks with no satellite input at all. The regional default rate is the most accurate estimator at both scales (block-mean MAE 0.98 vs 2.05 for no change), and S3 learns little more than that rate. Such a rate is in effect a local Tier 1 removal factor of the kind compiled by Requena Suarez *et al.* [100]. At this plot size and with these sensors, the information about growth comes from nearby field remeasurements, not from the pixels. Adding the Earth Engine inputs did not change this. These were the 2010 PALSAR mosaic, which makes the L-band record consistent at both censuses, CCI biomass change, and Hansen forest loss between the censuses. A change model on them lost to the regional default rate (block MAE 1.20 vs 0.98), and Hansen loss separated plots that lost biomass only weakly (AUC 0.57).

**Out-of-region stock on Borneo.** The same E2 model, trained without any Bornean plot, misses SAFE stock at $t_0$ by 132% rRMSE with a +80% bias (excluding oil palm). It over-predicts the 0.06 ha logged and riparian plots (field mean 99, predicted 217 $\text{Mg ha}^{-1}$), a second instance of the small-plot problem seen on HCSA.

TABLE XX
CREDITING RULES AT TARGET $\delta = 0.10$: REALISED OVER-CREDITING AND YIELD (31 PROJECTS, CROSS-REGION CALIBRATION)

| | **Over-credited** | | | |
|---|---|---|---|---|
| **Rule** | **All** | **Large** | **Small** | **Yield** |
| *Satellite only (no plot in the project)* | | | | |
| No deduction | 35.5% | 39.1 | 25.0 | 94.4% |
| VM0047, two-component $U$ | 0.0% | 0.0 | 0.0 | 47.8% |
| Parametric, two-component | 0.0% | 0.0 | 0.0 | 59.5% |
| Parametric, independent | 9.7% | – | – | 72.6% |
| Conformal (Eq. 16) | 9.7% | 13.0 | 0.0 | 72.2% |
| *Satellite + 5 random field plots (ratio estimator)* | | | | |
| VM0047 ($U - 10\%$) | 22.6% | 27.6 | 8.1 | 82.2% |
| Design lower bound | 9.9% | 10.2 | 9.2 | 80.7% |
| Conformal, plain | 9.4% | 3.8 | 25.3 | 80.8% |
| Conformal, studentized | 9.6% | 9.2 | 10.6 | 80.7% |
| *Satellite + 10 random field plots (16 projects)* | | | | |
| VM0047 ($U - 10\%$) | 29.3% | 42.1 | 12.9 | 90.0% |
| Design lower bound | 8.5% | 9.4 | 7.4 | 86.1% |
| Conformal, studentized | 6.5% | 6.1 | 7.0 | 82.1% |

Large: 1 ha and network plots (23 projects); small: 0.05 ha HCSA plots (8). With field plots: mean over 200 draws.

### J. From Intervals to Credits (RQ5)

Table XX credits the 31 projects under each rule at $\delta = 0.10$.

**Satellite only.** Crediting the raw map over-credits 35% of projects. The VM0047 deduction applied to the honest two-component uncertainty never over-credits, but it credits only 48% of the true stock. Conformal crediting meets its target: 9.7% of projects are over-credited (exact 95% interval 2–26% with 31 projects), and 72% of the true stock is credited without a single field plot in the project.

**With field plots.** The VM0047 deduction over-credits 23% of projects with 5 plots and 29% with 10, up to 42% of the large-plot projects. The design lower bound at the same $\delta$ realises 8.5–9.9% and gives up only 1.5–4 points of yield. Plain conformal calibration meets the target on average, but not by plot size: with 5 plots, it over-credits 25% of small-plot projects and 4% of large-plot ones. Studentization repairs this (10.6% and 9.2%), matching the design bound, whose validity comes from random sampling. Once plots are drawn at random, the classical design bound is the efficient choice, and conformal calibration adds a check, not credits.

**Is the bound calibrated whatever map it is given?** The satellite-only result above could be luck, because E2 under-predicts by about 9% on average and so rarely over-credits. Table XXI applies the same rules to five maps: E2, the Chloris product, their average, and E2 inflated by 10% and 20%, as an optimistic vendor map might be. Over the five maps, the conformal bound's realised over-crediting stays within 2.9 points of the target at $\delta = 0.10$, and within 2.6 points at $\delta = 0.20$. The independent parametric bound overshoots by up to 9.4 and 15.5 points. The two-component bound is conservative on E2 and drifts upward as the map is inflated.

TABLE XXI
SATELLITE-ONLY BOUNDS APPLIED TO FIVE MAPS ($\delta = 0.10$): REALISED OVER-CREDITING / YIELD

| Map | Param., indep. | Param., two-comp. | Conformal |
|---|---|---|---|
| E2 (blind LightGBM) | 9.7 / 72.6 | 0.0 / 59.5 | 9.7 / 72.2 |
| Chloris product | 19.4 / 77.6 | 3.2 / 53.7 | 12.9 / 66.2 |
| E2–Chloris average | 6.5 / 75.5 | 0.0 / 60.8 | 9.7 / 79.3 |
| E2 × 1.1 | 16.1 / 79.8 | 3.2 / 65.4 | 9.7 / 72.2 |
| E2 × 1.2 | 19.4 / 87.1 | 6.5 / 71.3 | 9.7 / 72.2 |
| Worst excess over $\delta$ | +9.4 | −3.5 | +2.9 |

Percent of projects over-credited / percent of true stock credited.

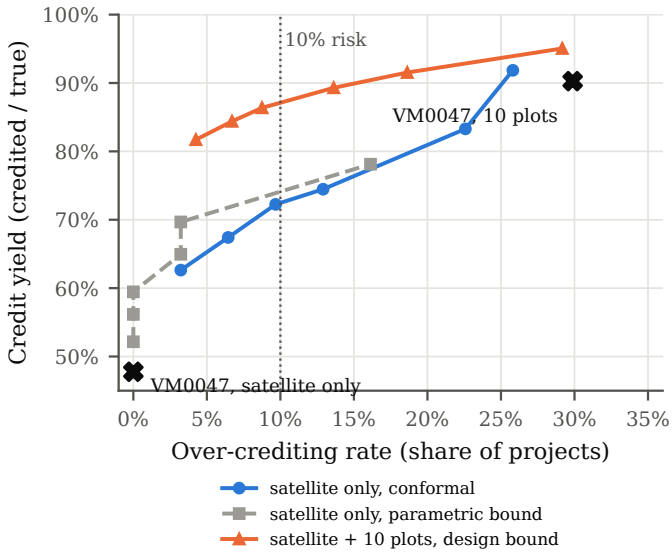



Fig. 10. Integrity-yield frontier: realised over-crediting against credit yield as the target $\delta$ varies from 0.05 to 0.30. Crosses mark the VM0047 deduction.

Inflation also shows the incentive. Under the parametric bound, a map inflated by 20% earns 87% of the true stock instead of 73%, and doubles the over-crediting rate. Under conformal crediting it earns exactly the same 72.2%, as the invariance property predicts. With the sharper map of Table VI (canopy height and GEDI), conformal crediting met the same target (9.7%) and credited 84% of the true stock instead of 72%. A better map buys credits; it does not buy risk.

**Integrity-yield frontier.** Fig. 10 traces each rule as $\delta$ varies. Both VM0047 points lie below the frontier. With 10 plots, VM0047 takes 29% risk for 90% yield, where the design bound gives 92% at 19%. Satellite-only, it credits 48% at no realised risk, where the other bounds give 60–72% at 0–10%. The frontier also prices a field plot. At about 10% realised risk, the satellite alone earns 72% of the true stock, 5 random plots raise it to 81%, and 10 plots to 86%. The parametric two-component bound lies slightly above the conformal curve at low risk. It is not better calibrated: its realised risk sits far from its nominal one, in a direction that changes with the map (Table XXI). A user of the parametric bound cannot know beforehand what risk is being taken.

**What the guarantee needs.** The guarantee is marginal over the population of projects in the calibration bank, and it is only as good as that bank. With a mixed bank, network-plot projects were over-credited in 33% of cases at $\delta = 0.10$, against 0% for the 1 ha and HCSA projects. Calibrating only on projects from other datasets raised this to 44%. Calibrating only on the same dataset brought it to 0–14%, but with 6–8 calibration projects the quantile reached the largest score, and the mean yield fell to 18%. A conditional guarantee for a project type therefore needs at least 9 field-verified projects of that type for $\delta = 0.10$, and 19 for $\delta = 0.05$. Two further limits apply. The truth in the bank is itself a field estimate, so the guarantee is relative to that reference. And projects with field truth may not be selected like the projects being credited, which is the main threat to exchangeability.

**Position.** The conservativeness target is that of Grassi *et al.* and Neeff [86], [87], and the machinery is split and normalized conformal prediction [43], [44]. Conformal intervals for area-level parameters have been proposed for small-area estimation [101], and for the true effect of a new trial in meta-analysis [102]. What we have not found before is the following: a crediting rule calibrated across projects against field truth, its empirical test under spatial transfer and biased maps, its invariance to map inflation, and a measured integrity-yield trade-off for the registry deduction.

### K. How Reliable Are These Numbers?

Every score above comes from 27 region blocks, and errors are correlated within a block. We therefore resampled whole blocks with replacement (2,000 draws) and refitted E2 with five LightGBM seeds (Table XXII). Seed noise is small: the overall rRMSE moved by 0.36 points (SD). Region sampling is not: the 95% interval of the overall rRMSE runs from 35% to 69%. Three kinds of claim survive this test. The first is the differences between datasets and plot sizes, whose intervals do not overlap. The second is the gain of the E2–Chloris average over E2 ($-3.7$ points, interval $-7.2$ to $-0.6$). The third is the coverage gain of cloud-aware calibration under full cloud ($+4.0$ points, $+1.9$ to $+6.6$). Two claims do not survive. The difference between E2 and Chloris ($-0.8$ points, $-8.3$ to $+5.3$) and between radar and optical inputs ($-2.3$, $-11.1$ to $+5.1$) are within noise. We report them as tendencies, not findings.

### L. Systems (RQ6)

**Production runs.** The stock extraction ran 2,052 tasks on 24 workers in 2.1 h (0.26 tasks $s^{-1}$) with a measured load factor of 1.47, close to the 1.40 predicted from task counts; 85 transient errors were absorbed by retries. We stopped the first attempt, which used tile-only placement, to switch policies; the restart skipped the 92 tasks already in the content-addressed cache. Seventeen tasks (0.8%) failed permanently: 13 hit two Landsat scenes whose files could not be read from the catalogue, and four hit unreadable Sentinel-2 files. Retrying cannot repair a permanently broken input; the runner records such tasks and the model treats their features as missing, and a scene-level fallback is the obvious next step. The SAFE extraction ran

TABLE XXII
RELIABILITY: REGION-BLOCK BOOTSTRAP 95% INTERVALS (2,000 DRAWS) AND SEED SPREAD

| Quantity | Value | 95% interval |
|---|---|---|
| rRMSE E2, all plots | 48.5% | 34.9–68.7 |
| rRMSE E2, Labrière (1 ha) | 28.6% | 22.8–38.4 |
| rRMSE E2, GEO-TREES | 41.9% | 35.9–51.8 |
| rRMSE E2, HCSA | 102.9% | 88.1–121.7 |
| E2 − Chloris (points) | −0.8 | −8.3 to +5.3 |
| E2–Chloris average − E2 (points) | −3.7 | −7.2 to −0.6 |
| Radar only − optical only (points) | −2.3 | −11.1 to +5.1 |
| CQR coverage, as observed | 89.9% | 83.9–94.4 |
| Cloud-aware CQR coverage, fully clouded | 90.5% | 84.7–95.2 |
| Cloud-aware − plain, fully clouded (points) | +4.0 | +1.9 to +6.6 |
| E2 rRMSE over 5 seeds (mean ± SD) | 48.8 | ±0.36 |

TABLE XXIII
STRONG SCALING AND FAULT INJECTION (48 TASKS, EMPTY CACHE)

| Workers | Fault $p$ | Wall (s) | Speed-up | $\lambda$ | Identical |
|---|---|---|---|---|---|
| 1 | 0 | 498.5 | 1.00 | 1.00 | 47/47 |
| 2 | 0 | 207.8 | 2.40 | 1.00 | 47/47 |
| 4 | 0 | 144.8 | 3.44 | 1.00 | 47/47 |
| 8 | 0 | 98.4 | 5.06 | 1.00 | 47/47 |
| 16 | 0 | 93.3 | 5.34 | 1.33 | 47/47 |
| 32 | 0 | 91.4 | 5.45 | 2.67 | 47/47 |
| 8 | 0.1 | 106.7 | 4.67 | 1.00 | 47/47 |
| 8 | 0.2 | 105.0 | 4.75 | 1.00 | 47/47 |
| 8 | 0.3 | 105.7 | 4.72 | 1.00 | 47/47 |

2,006 tasks on 16 workers in 2.3 h with no permanent failures, 36 retries, and a load factor of 1.44.

**Scaling, faults, and resume.** A random sample of 48 tasks (Landsat, Sentinel-1, PALSAR, terrain) was run on 1 to 32 workers, each time on an empty cache (Table XXIII). Throughput rose 5.1-fold up to 8 workers and then flattened (5.5-fold at 32). Two limits explain this. With 48 tasks, 32 workers hold one or two buckets each, so the load factor rose to 2.67. And the makespan cannot fall below the slowest single task, a multi-scene read, however many workers wait beside it. The extraction is bound by remote reads, not by local compute. Every run was compared with the single-worker run task by task: all 47 finished results were bit-identical at every worker count. With transient faults injected at rates 0.1, 0.2, and 0.3 per attempt, the runner absorbed 13, 18, and 28 retries, and again all 47 results matched. The one task that never finished, in any run, is the one whose Landsat file is unreadable, which is a permanent fault. In the resume test, a run stopped after 23 tasks; the restart skipped those 23 from the cache, ran the remaining 24, and produced output identical to the uninterrupted run.

The super-linear step from one to two workers reflects variable latency of the remote service between runs; each configuration was run once, so differences of a few seconds are within noise.

**Ledger.** The final log holds 37 entries : four seals (one per right-grid build and one per evaluation run), three commits, two reveals, and 28 label releases, none of which included a label from its target block. The chain verifies end to end. In unit tests, editing any entry broke verification, a commit after a reveal was refused, and a reveal with a modified outcome file was refused.

## V. DISCUSSION

### A. Can Satellites Replace the Field Plot?

Not at the scale of a plot, and not for growth. Tested blind, open satellite data put a 1 ha plot's carbon stock within roughly 20–30% of the field-formula value once canopy height and GEDI are included, and a 0.05 ha plot within about 100%; the best honest interval (CQR) is a factor of five wide. For sequestration, the quantity on which credits are paid, no satellite estimator we built beat assuming no change, and a regional average of measured growth beat all of them. A platform that issues plot-level or annual growth numbers from these sensors alone is reporting a default constant, whether or not it says so.

Nor can a satellite count trees. The field ground truth is a list of stems with diameters, heights, and species. The 1 m canopy height map shows the top of the canopy: it found 3% of the stems, cannot see the 75% of stems below 15 cm or anything under the crowns, and cannot see species, which alone moves a plot's carbon by a fifth. Applying the field formula to what the map shows finds a quarter of the carbon. The satellite's real "fixed constant" is canopy height, not a tree count, and the step from height to carbon has to be learned from field plots.

What satellites do well is replace part of the inventory. Used as auxiliary data in a ratio estimator, they were 1.4–2.3 times as efficient as field plots alone on large-plot datasets, roughly halving the plots needed for a given precision (Table XVIII). The useful framing is not "satellites instead of field plots" but "satellites plus a small, well-designed set of field plots instead of a full inventory". Even that only holds where the satellite can rank plots within a project. On the small Indonesian plots it cannot (within-region correlation 0.2), and there it does not save a single plot.

### B. Our Numbers Against the Literature

Table XXIV places our blind results next to published validations. Our 1 ha error (29%) is close to the 31–38% reported for Sentinel-1 and PALSAR in Kalimantan [103]. That study was validated against a lidar-upscaled reference layer, not blind against field plots, and it reports the same underestimation above 200 Mg ha$^{-1}$. Our saturation near 300 Mg ha$^{-1}$ matches the flattening near 250 Mg ha$^{-1}$ in the global map of Santoro *et al.* [4] and the steep rise of map error above 300 Mg ha$^{-1}$ found by Araza *et al.* [14]. Airborne lidar in Sabah, the best remote sensing can do, still errs by 25% at 0.1 ha and 13% at 1 ha [94]. Plot size dominates every comparison. Local variation of AGB alone has a coefficient of variation of 46% at 0.1 ha and 17% at 1 ha [13], and Duncanson *et al.* recommend plots of at least 1 ha for validation [51]. If we remove that expected representativeness noise in quadrature, the blind satellite error is about 23% on the 1 ha units and

TABLE XXIV
BLIND RESULTS OF THIS STUDY AGAINST PUBLISHED BENCHMARKS

| Study | Sensor | Reference | Error |
|---|---|---|---|
| This study, Labrière (blind) | multi-sensor | 1 ha plots | 28.6% |
| This study, + canopy ht. + GEDI | + 1 m CHM, GEDI | 1 ha plots | 20.6% |
| This study, GEO-TREES (blind) | same | network | 41.9% |
| This study, HCSA (blind) | same | 0.05 ha plots | 102.9% |
| Berninger 2018, Kalimantan [103] | S1 + PALSAR | lidar layer | 31–38% |
| Jucker 2018, Sabah [94] | airborne lidar | 0.1 / 1 ha | 25 / 13% |
| Araza 2022, CCI [14] | CCI map | $0.1^\circ$ cells | 29–61* |
| Santoro 2021 [4] | GlobBiomass | 110 k plots | SD ≈50% |

*RMSD in $\mathrm{Mg\,ha^{-1}}$ for AGB below 300; 107 at 300–400. Other errors are rRMSE.

still about 90% on HCSA. Plot size alone does not explain the Indonesian result. The mosaic of open land, scrub, and forest around HCSA plots, and a GPS error comparable to the 12.6 m plot radius [51], are the likely remainder.

### C. Clouds: What Actually Matters

Three findings change how clouds should be handled. First, *how often* a plot is clear matters less than *how* clearness is decided. A clear look within $\pm 30$ days exists for 96% of dates in Indonesia, because revisit compensates for an 80% cloud frequency [32], [34]. The scene-level filter that most pipelines apply, by contrast, discards two thirds of the clear looks and keeps a third of cloudy ones. Pixel-level masking at the plot, with SCL, Fmask, or Cloud Score+ [35]–[37], nearly triples the usable data. Second, *season matters*: in Indonesia the clear share in May is 4.4 times that in January, so field campaigns and their satellite windows should be planned for the dry months. Third, a model led by radar and an annual product loses only two points of accuracy without any optical data, so clouds are an availability problem for optical indices but not for a well-designed multi-sensor stock estimate. That is the practical form of the SAR–optical fusion argued for by Reiche *et al.* [41]. Filling optical gaps from radar and feeding them to the model, however, was the worst option we tested. Reconstructed reflectance is a model output whose error does not disappear because the image looks complete.

### D. Confidence That Means Something

Honest confidence has three layers, and each failed in a characteristic way. At *plot* level, conformal intervals were honest overall, but only CQR was both honest and usable, and missing-data-augmented calibration [8] kept it honest when optical data were missing, outside the assumptions under which that method is guaranteed. The guarantee is marginal: under the shift from large to small plots (HCSA), coverage fell to 68–81% for every method. Spatially weighted conformal prediction [46], [47] is the next step. At *project* level, independence is the dangerous assumption. Map errors are correlated over tens of kilometres [14], a project shares one regional bias, and an interval that ignores it covered only 74% of projects. The two-component interval is honest, and it shows that a satellite-only project estimate would lose most of its credits under VM0047. At the level of the *standard*, a deduction of $U - 10\%$ is not a lower confidence bound. With ten field plots, the deducted estimate still exceeded the true stock in a third of draws. Conformal crediting (Section IV-J) makes the over-crediting rate a design parameter rather than an accident, and removes the reward for inflating the map.

A metadata quality score answers a different question from a confidence interval. Ours rewarded recent, cloud-checked imagery. Over all plots it trusted most the plots the model got most wrong, and within a dataset it said nothing at all about error. Confidence has to be learned from errors against field data and reported in the units of the claim, as an interval in $\mathrm{t\,C\,ha^{-1}}$, not as a grade.

### E. The Ground Truth Is a Model Too

The right grid is fixed by the standards, but the standards leave real choices, and those choices are as large as the precision targets they enforce. Switching the root-to-shoot ratio from the SNI 7724 value (IPCC 2006) to the 2019 Refinement changes total carbon by 13%. Switching from the pantropical Chave equation to the East-Kalimantan equation of Basuki *et al.* leaves the national total unchanged but moves provincial stocks from $-40\%$ to $+121\%$. AR-TOOL14 counts sampling error only, yet measurement and allometry carried up to 64% of the variance of a provincial mean. The same data receive deductions from 0 to 34% depending on which standard is applied. For growth the field answer itself is fragile. With 9–16 plots of 0.06 ha, VM0047 Eq. 28 deducts most or all of the credit unless the correlation of permanent plots is used, and the plot-level measurement error of the change halves only when the same stem is assessed with the same equation at both dates. The target "100% accuracy" therefore has no meaning. The meaningful target is to fall within the uncertainty of a field estimate made to a stated standard, and that uncertainty must itself be computed in full.

### F. The Path Toward the Field Answer

The experiments rank the levers. (i) *Report at project scale*: aggregation removes most random error. (ii) *Add local plots and use the satellite as auxiliary data*, which is the most effective step, provided the plots are large enough to be ranked by the satellite. (iii) *Mask clouds at the plot, not the scene*, and match the window to the season. (iv) *Lead with structure-sensitive sensors*. Radar was ahead of optical on both large-plot datasets, though within sampling noise, and it is what keeps the estimate alive under cloud; adding a 1 m canopy height map and GEDI cut the blind error by 9 points (Table VI), and drone lidar (Section V-G) and the 2025 L- and P-band missions NISAR and ESA BIOMASS are the next steps. (v) *Keep sensors consistent through time*: half of the sequestration error came from Sentinel-1 appearing between the two censuses. (vi) *Combine rather than choose*: averaging the transfer model with an independent global product was the best blind stock estimate. (vii) *Calibrate confidence*: CQR with cloud-aware calibration at plot level, the two-component variance at project level, and credit the lower bound.

### G. Toward a Multi-Tier System: Drones

Three results of this study point in one direction. The biggest gain in accuracy came from fine-scale canopy structure: a 1 m canopy height map. Plots smaller than about 0.25 ha cannot be predicted by anything we tried. Trees cannot be counted on a satellite canopy height map (Section IV-E). And the most effective step toward the field answer is more field plots. NusaCarbon's planned drones target all three. We set out the design and what each tier still has to prove; no drone data enter the results above.

**Under the canopy.** The swarm replaces the slowest step of the right grid: measuring every stem's diameter. Under-canopy laser scanning from a piloted drone has reached a DBH error of 0.6–0.9 cm (2–3%) in boreal plots [104]. Fully autonomous flight reached 3.8% in a simple stand but 20–22% in complex ones [105]. Swarms of micro drones already navigate dense forest autonomously [106], but none carries a measuring payload yet, and we found no under-canopy DBH study in tropical forest. Species, needed for wood density, can fall back on genus means, which carry 74% of species-level variation [108]. Drone imagery has classified key restoration species in Indonesia with 74–92% accuracy [109]. Plugged into Step 5 of the right grid, the swarm's own measured DBH error replaces the Chave 2004 error model, so its uncertainty is propagated rather than assumed.

**Above the canopy.** The Zenmuse L2 gives a canopy height model at centimetre scale over about 2.5 $km^2$ per flight, according to the manufacturer. We found no peer-reviewed test under tropical canopy. Airborne lidar in Sabah reached a plot error of 13% at 1 ha, against far larger errors for satellite maps at the same plots [94]. In Peru its uncertainty fell to about 5% at 5 ha [110]. A chain from field plots to drone lidar to Sentinel-2 has beaten direct field-to-satellite models, and it needed about 37% fewer field plots [111]. IPCC accepts biomass maps built from field and remote-sensing data at Tiers 2 and 3 [112]. A repeat flight measures canopy-height change on the same trees, which is the quantity satellites failed to resolve here.

**How to test it.** The protocol of this paper applies unchanged. Under-canopy plots, computed by the standard chain, are the sealed right grid. Drone-lidar and satellite estimates are the left grid. The ledger, conformal intervals, and conformal crediting carry over. Three hypotheses should be tested on Indonesian 1 ha plots before any claim is made. (H1) Swarm DBH is within 3% of tape on tropical stems, with at least 90% of stems detected. (H2) L2-based AGB is within 15% of the right grid at 1 ha, leave-one-site-out. (H3) Repeat-flight growth beats both zero change and the regional default rate at block level. If H2 holds, the satellite tier can be recalibrated on drone-lidar maps instead of sparse plots, which is how GEDI's own under-represented Southeast-Asian strata could be filled [25].

### H. Implications for NusaCarbon

Our audit found default constants where measurements were expected, a scene-level cloud filter, and a confidence grade computed from image metadata. The evidence suggests a concrete redesign:

- Mask clouds per pixel at the plot (SCL or Cloud Score+) instead of `CLOUDY_PIXEL_PERCENTAGE`, and widen the window in the wet season.
- Estimate stock with the production method of Section IV-G: the embeddings map with canopy height and GEDI, corrected by the realm's regional bias, with a cloud-aware CQR interval instead of the letter grade.
- Report projects with the two-component interval, and calibrate each project with ten to twenty large field plots through a ratio estimator.
- Credit with a bound whose risk is calibrated: conformal crediting with one score per field-verified region when no plot is measured, the design lower bound when random plots are, never the $U - 10\%$ deduction alone.
- Credit sequestration from field remeasurement of permanent plots, and use satellites to detect loss and disturbance.
- Log every estimate through the commit-reveal ledger.
- Replace the height formula, which matches no published equation and converts carbon to carbon twice, and fix the GEDI band it reads, which does not exist.

### I. Threats to Validity

*Internal.*

- The right grid is itself a model: Chave allometry carries its own error, and GEO-TREES and Labrière AGB were computed by their authors, not by us.
- HCSA is released as a 2023 dataset without per-plot dates, and Labrière plots have campaign-year dates only, so temporal matching for those plots is at year precision.
- The standard formula failed on SAFE oil palm. Those plots are excluded by a subset fixed before the reveal, and 45 palm stems remain in forest plots.
- The improvement ladder, the change models, the cloud and interval experiments, the project-level analysis, E4, and the project simulations were defined after the reveals and are exploratory. Only the E0, E2, S0, and S2 scores and the Danum test are confirmatory, and we report every variant we ran; Section IV-K states which differences exceed sampling noise.
- The ledger proves the order of events, not the absence of leakage upstream of it. GEDI biomass models were calibrated on field plots and airborne lidar that may include some of our sites, notably in Gabon; the canopy-height maps were trained on lidar outside our plots, and most of the gain in Table VI comes from them. The Chloris product, used as E0 and as an input of the ensemble, was calibrated by its makers on field and lidar data that we cannot inspect; if any of our plots were among them, that path leaks. E2 without the product is free of this path.
- Provincial stratum weights are plot shares, because mapped stratum areas are not public.

TABLE XXV
PLANNED NUSACARBON TIERS: WHAT EACH MEASURES, PUBLISHED EVIDENCE, AND WHAT REMAINS TO BE SHOWN

| Tier | Sensor | Role | Published evidence | To be shown |
|---|---|---|---|---|
| Under canopy | drone swarm with laser scanner (to be built) | Measure $D$ and stem position on 1 ha plots without a field crew; becomes the right grid | Under-canopy UAV laser scanning: DBH RMSE 0.6–0.9 cm (2–3%), stems detected 84–93%, boreal, piloted [104]; autonomous flight 3.8% to 22% by stand complexity [105]; swarms navigate forest [106] but none has published DBH accuracy | DBH accuracy and stem detection under tropical canopy; full autonomy at 1 ha |
| Above canopy | DJI Matrice 350 RTK + Zenmuse L2 lidar | Canopy height over the whole project, converted to AGB with the under-canopy plots; repeat flights for change | L2: vertical accuracy 4 cm at 150 m, up to 5 returns, about 2.5 $km^2$ per flight (manufacturer); airborne lidar in Sabah: 13% plot error at 1 ha [94]; Kalimantan lidar–AGB $R^2 \approx 0.8$ [107] | AGB error at 1 ha in Indonesian forest; ground detection under dense canopy; growth from repeat flights |
| Satellite | Sentinel-1/2, Landsat, PALSAR, GEDI, 1 m canopy height | Wall-to-wall stock, yearly disturbance monitoring, screening; calibrated by the drone tiers | This study: 21% at 1 ha, 7–9% on project totals outside Indonesia, no skill for growth | Gain when calibrated by drone lidar instead of sparse plots |

- The cloud climatology uses SCL, whose own overall accuracy is about 80–90% [36].

*Construct.*

- Plot-scale truth (0.05–1 ha) is compared with 10 m to 4.6 km pixels, and GPS error on small plots adds noise that no estimator can remove.
- Conformal intervals calibrated on plots cover plot values, not true stocks.
- GEDI biomass models were calibrated on field plots elsewhere, so E1 is free of our field data but not of field data altogether.

*External.* The unseen test covers one valley of tall Bornean forest. Indonesian plots come from one programme, collected by trained non-foresters; results may not transfer to other forest types or survey teams. Projects in the project-level analysis are region blocks, not registered projects.

## VI. CONCLUSION

We asked whether open satellite data can replace the field plot in forest carbon MRV, and tested it blind. A sealed right grid was computed step by step to SNI 7724, IPCC, CDM, Verra, and Gold Standard rules with a full uncertainty budget. A left grid saw only coordinates and dates. A hash-chained ledger makes any change to a prediction after the answer was opened detectable.

Across 476 stock plots, the blind estimate missed by 49% relative RMSE overall with Sentinel, Landsat, and PALSAR, and by 40% after adding a 1 m canopy height map and GEDI lidar: 21% on 1 ha plots, 7–9% on project totals outside Indonesia, and still about 100% on small Indonesian plots. Sharper canopy structure, not a better algorithm, gave the largest gain, which is the case for drone lidar. Counting trees on the 1 m map and applying the field formula found 3% of the stems and a quarter of the carbon, and in 1,000 simulated projects the best map found a median 95% of the field carbon, within 25% in four projects of five; satellite embeddings and a realm bias correction raised this to 88.5%, and the frozen method found 85% of the field carbon on 35 unseen Bornean plots. In Indonesia only 21.5% of Sentinel-2 acquisitions see a plot clear. Masking at the plot instead of filtering scenes nearly triples the usable data, and a radar-led model loses only two points without optical data. Missing-data-augmented conformal calibration kept 90% intervals honest under full cloud, while a metadata confidence score carried no information about error beyond dataset differences. At project scale, an interval that ignores spatially correlated error covered only 74% of projects; with field plots as calibration, the satellite was 1.4–2.3 times as efficient as field plots alone on large-plot datasets. Conformal crediting met a stated over-crediting risk without a single field plot in the project, earning 72% of the true stock at 10% risk, was invariant to map inflation, and showed that the registry deduction sits below the integrity-yield frontier. For annual sequestration, no satellite estimator beat "no change", and even the field answer was too uncertain to credit without the correlation of permanent plots.

Satellites are a stock-mapping, disturbance, and plot-saving tool, not a replacement for the field plot, and their confidence must be calibrated, not graded. The contribution is the evaluation, not a new method: the protocol, the ledger, and the extraction runner are reusable. Together they turn a platform's accuracy claim into something a third party can check.

## APPENDIX A
## PER-PLOT ROWS OF THE TEN SIMULATED PROJECTS

Tables XXVI and XXVII list every plot of the ten simulations of Table X. Stems: field stems per hectare from the HCSA stem list ("–" where only plot biomass is published). Trees: trees per hectare that E4 counted on the 1 m canopy height map. Carbon columns are $t\,CO_2e\,ha^{-1}$; Map % is the map as a percentage of the field value. Rows for all 475 plots are in `tree_benchmark_plots.csv` and in the dashboard.

TABLE XXVI
PER-PLOT ROWS, SIMULATIONS 1–5

| Sim | Region | Plot | Stems | Trees | Field | Map | Map % | E4 | E4 sc. |
|---|---|---|---|---|---|---|---|---|---|
| 1 | HCSA Riau | HCSA_Riau_RU05 | 960 | 0 | 462 | 237 | 51 | 0 | 0 |
| | | HCSA_Riau_RU21 | 500 | 0 | 10 | 63 | 611 | 0 | 0 |
| | | HCSA_Riau_RU06 | 720 | 0 | 170 | 120 | 71 | 0 | 0 |
| | | HCSA_Riau_RU15 | 580 | 20 | 114 | 463 | 407 | 30 | 117 |
| | | HCSA_Riau_RU01 | 1960 | 0 | 833 | 152 | 18 | 0 | 0 |
| | | HCSA_Riau_RU22 | 60 | 40 | 12 | 78 | 649 | 23 | 90 |
| | | HCSA_Riau_RU10 | 220 | 40 | 79 | 87 | 111 | 29 | 113 |
| | | HCSA_Riau_RU19 | 420 | 0 | 45 | 91 | 203 | 0 | 0 |
| | | HCSA_Riau_RU14 | 120 | 20 | 49 | 204 | 419 | 194 | 745 |
| | | HCSA_Riau_RU03 | 160 | 20 | 602 | 181 | 30 | 218 | 839 |
| 2 | Labrière Nouragues | LAB_NOU04h5 | – | 15 | 719 | 789 | 110 | 168 | 631 |
| | | LAB_NOU02h10 | – | 23 | 748 | 755 | 101 | 201 | 752 |
| | | LAB_NOU03h2 | – | 15 | 893 | 627 | 70 | 126 | 474 |
| | | LAB_NOU04h9 | – | 16 | 876 | 719 | 82 | 175 | 656 |
| | | LAB_NOU02h6 | – | 22 | 532 | 690 | 130 | 149 | 561 |
| | | LAB_NOU02h5 | – | 18 | 497 | 637 | 128 | 139 | 522 |
| | | LAB_NOU04h12 | – | 19 | 805 | 657 | 82 | 183 | 687 |
| | | LAB_NOU02h1 | – | 19 | 490 | 560 | 114 | 106 | 397 |
| | | LAB_NOU04h2 | – | 24 | 508 | 678 | 134 | 218 | 818 |
| | | LAB_NOU02h9 | – | 18 | 795 | 712 | 90 | 142 | 531 |
| 3 | Labrière Mondah | LAB_MON11h | – | 23 | 60 | 59 | 98 | 6 | 22 |
| | | LAB_MON05h | – | 18 | 161 | 167 | 104 | 13 | 50 |
| | | LAB_MON22h | – | 9 | 498 | 453 | 91 | 32 | 122 |
| | | LAB_MON01h | – | 20 | 44 | 49 | 112 | 4 | 15 |
| | | LAB_MON21Ah | – | 27 | 276 | 236 | 85 | 54 | 202 |
| | | LAB_MNG03h | – | 29 | 880 | 496 | 56 | 164 | 617 |
| | | LAB_MNG04h | – | 21 | 765 | 559 | 73 | 152 | 574 |
| | | LAB_MON14h | – | 23 | 258 | 281 | 109 | 41 | 154 |
| | | LAB_MON19h | – | 3 | 6 | 9 | 145 | 0 | 1 |
| | | LAB_MON13h | – | 17 | 427 | 534 | 125 | 74 | 281 |
| 4 | Labrière Mabounie | LAB_MAB05h | – | 9 | 756 | 717 | 95 | 106 | 400 |
| | | LAB_MAB11h | – | 18 | 902 | 701 | 78 | 166 | 625 |
| | | LAB_MAB03h | – | 0 | 575 | 187 | 33 | 0 | 0 |
| | | LAB_MAB10h | – | 4 | 594 | 200 | 34 | 11 | 40 |
| | | LAB_MAB12h | – | 16 | 296 | 544 | 184 | 121 | 456 |
| | | LAB_MAB04h | – | 18 | 791 | 659 | 83 | 209 | 788 |
| | | LAB_MAB07h | – | 12 | 426 | 550 | 129 | 88 | 334 |
| | | LAB_MAB01h | – | 23 | 565 | 489 | 87 | 134 | 505 |
| | | LAB_MAB08h | – | 22 | 501 | 576 | 115 | 150 | 566 |
| | | LAB_MAB06h | – | 14 | 533 | 548 | 103 | 129 | 488 |
| 5 | HCSA West Papua | HCSA_West_Papua_PB12 | 580 | 0 | 89 | 108 | 121 | 0 | 0 |
| | | HCSA_West_Papua_PB01 | 860 | 0 | 192 | 220 | 114 | 0 | 0 |
| | | HCSA_West_Papua_PB19 | 1260 | 0 | 556 | 125 | 22 | 0 | 0 |
| | | HCSA_West_Papua_PB09 | 600 | 20 | 532 | 301 | 57 | 399 | 1546 |
| | | HCSA_West_Papua_PB20 | 1340 | 0 | 224 | 356 | 158 | 0 | 0 |
| | | HCSA_West_Papua_PB05 | 500 | 20 | 75 | 106 | 142 | 20 | 79 |
| | | HCSA_West_Papua_PB11 | 900 | 20 | 376 | 302 | 80 | 181 | 700 |
| | | HCSA_West_Papua_PB21 | 1320 | 20 | 190 | 304 | 160 | 206 | 798 |
| | | HCSA_West_Papua_PB17 | 340 | 60 | 83 | 272 | 328 | 238 | 923 |
| | | HCSA_West_Papua_PB07 | 1540 | 20 | 225 | 229 | 102 | 138 | 534 |

TABLE XXVII
PER-PLOT ROWS, SIMULATIONS 6–10

| Sim | Region | Plot | Stems | Trees | Field | Map | Map % | E4 | E4 sc. |
|---|---|---|---|---|---|---|---|---|---|
| 6 | HCSA Riau | HCSA_Riau_RU16 | 80 | 20 | 146 | 148 | 102 | 74 | 283 |
| | | HCSA_Riau_RU06 | 720 | 0 | 170 | 120 | 71 | 0 | 0 |
| | | HCSA_Riau_RU04 | 1120 | 0 | 202 | 159 | 79 | 0 | 0 |
| | | HCSA_Riau_RU05 | 960 | 0 | 462 | 237 | 51 | 0 | 0 |
| | | HCSA_Riau_RU22 | 60 | 40 | 12 | 78 | 649 | 23 | 90 |
| | | HCSA_Riau_RU12 | 500 | 0 | 195 | 259 | 133 | 0 | 0 |
| | | HCSA_Riau_RU08 | 720 | 80 | 460 | 119 | 26 | 30 | 114 |
| | | HCSA_Riau_RU11 | 820 | 0 | 104 | 156 | 150 | 0 | 0 |
| | | HCSA_Riau_RU15 | 580 | 20 | 114 | 463 | 407 | 30 | 117 |
| | | HCSA_Riau_RU19 | 420 | 0 | 45 | 91 | 203 | 0 | 0 |
| 7 | Labrière Rabi | LAB_RAB12h | – | 23 | 534 | 588 | 110 | 213 | 816 |
| | | LAB_RAB17h | – | 14 | 458 | 500 | 109 | 175 | 672 |
| | | LAB_RAB18h | – | 11 | 452 | 581 | 128 | 128 | 492 |
| | | LAB_RAB05h | – | 19 | 553 | 469 | 85 | 128 | 491 |
| | | LAB_RAB02h | – | 19 | 483 | 501 | 104 | 146 | 559 |
| | | LAB_RAB16h | – | 24 | 437 | 471 | 108 | 151 | 577 |
| | | LAB_RAB19h | – | 22 | 440 | 423 | 96 | 151 | 581 |
| | | LAB_RAB11h | – | 15 | 496 | 483 | 97 | 154 | 589 |
| | | LAB_RAB25h | – | 18 | 506 | 522 | 103 | 157 | 604 |
| | | LAB_RAB08h | – | 19 | 361 | 504 | 140 | 148 | 566 |
| 8 | Labrière Mabounie | LAB_MAB10h | – | 4 | 594 | 200 | 34 | 11 | 40 |
| | | LAB_MAB07h | – | 12 | 426 | 550 | 129 | 88 | 334 |
| | | LAB_MAB04h | – | 18 | 791 | 659 | 83 | 209 | 788 |
| | | LAB_MAB03h | – | 0 | 575 | 187 | 33 | 0 | 0 |
| | | LAB_MAB01h | – | 23 | 565 | 489 | 87 | 134 | 505 |
| | | LAB_MAB08h | – | 22 | 501 | 576 | 115 | 150 | 566 |
| | | LAB_MAB05h | – | 9 | 756 | 717 | 95 | 106 | 400 |
| | | LAB_MAB12h | – | 16 | 296 | 544 | 184 | 121 | 456 |
| | | LAB_MAB06h | – | 14 | 533 | 548 | 103 | 129 | 488 |
| | | LAB_MAB02h | – | 12 | 521 | 624 | 120 | 112 | 422 |
| 9 | HCSA Jambi | HCSA_Jambi_JB15 | 540 | 60 | 54 | 106 | 194 | 10 | 39 |
| | | HCSA_Jambi_JB02 | 960 | 20 | 180 | 298 | 165 | 145 | 558 |
| | | HCSA_Jambi_JB05 | 1300 | 20 | 180 | 538 | 298 | 232 | 893 |
| | | HCSA_Jambi_JB19 | 560 | 0 | 28 | 84 | 304 | 0 | 0 |
| | | HCSA_Jambi_JB12 | 1180 | 0 | 291 | 261 | 90 | 0 | 0 |
| | | HCSA_Jambi_JB27 | 340 | 20 | 719 | 189 | 26 | 133 | 511 |
| | | HCSA_Jambi_JB14 | 1420 | 20 | 294 | 116 | 39 | 4 | 16 |
| | | HCSA_Jambi_JB18 | 580 | 40 | 222 | 553 | 249 | 132 | 508 |
| | | HCSA_Jambi_JB28 | 160 | 0 | 613 | 117 | 19 | 0 | 0 |
| | | HCSA_Jambi_JB30 | 380 | 0 | 292 | 409 | 140 | 0 | 0 |
| 10 | Labrière Mondah | LAB_MON19h | – | 3 | 6 | 9 | 145 | 0 | 1 |
| | | LAB_MON21h | – | 3 | 4 | 6 | 146 | 0 | 1 |
| | | LAB_MON22h | – | 9 | 498 | 453 | 91 | 32 | 122 |
| | | LAB_MON23h | – | 22 | 222 | 288 | 130 | 49 | 185 |
| | | LAB_MON10h | – | 14 | 178 | 212 | 119 | 10 | 37 |
| | | LAB_MON20h | – | 14 | 121 | 142 | 117 | 11 | 41 |
| | | LAB_MON13h | – | 17 | 427 | 534 | 125 | 74 | 281 |
| | | LAB_MON09h | – | 0 | 4 | 5 | 133 | 0 | 0 |
| | | LAB_MON11h | – | 23 | 60 | 59 | 98 | 6 | 22 |
| | | LAB_MON01h | – | 20 | 44 | 49 | 112 | 4 | 15 |

## Data and Code Availability

All field sources are public: HCSA Indonesia (CC BY-SA 4.0) [60], GEO-TREES [61], Labrière *et al.* (CC0) [62], and the SAFE tree census [89]. Satellite data were read from the Microsoft Planetary Computer [59] and Google Earth Engine. The pipeline (Python; pandas, LightGBM, rasterio, pystac) is in the project repository, including the ledger, the runner, the right-grid builder, and the evaluation scripts. Every reported number is regenerated by the modules `right_grid`, `right_grid_seq`, `right_grid_uncertainty`, `extract_left`, `evaluate`, `evaluate_seq`, `improve`, `improve_seq`, `window_sensitivity`, `cloud_climatology`, `cloud_analysis`, `robust`, `project_uncertainty`, `systems_experiments`, together with `extract_gee`, `accuracy_gee`, `conformal_crediting`, `crediting_frontier`, `crediting_robustness`, `extract_chm`, `tree_count`, `tree_benchmark`, `simulations`, `gain_loss`, `reliability`, and `figures`, of the `estimation` package. Wood density is from the GWDD copy shipped with BIOMASS v2.1.11 [20].

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